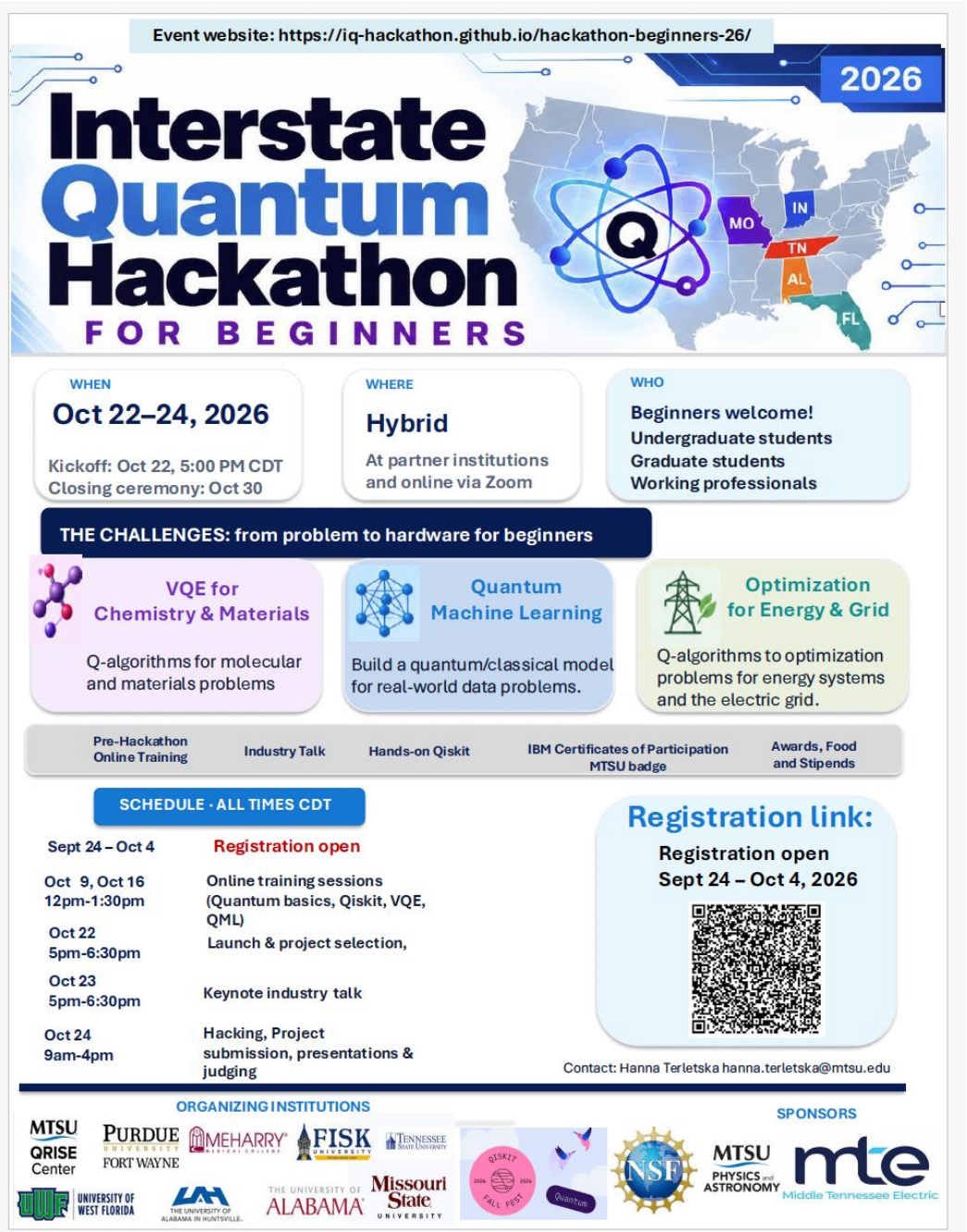


Pre-Hackathon Training
Introduction to Quantum Computing – part 1

Hanna Terletska

Professor, QRISE Director

Middle Tennessee State University

MTSU QUANTUM

RESEARCH - EDUCATION - COMPUTING - WORKFORCE

This lecture is for the VERY beginners to make sure we are al on the same track

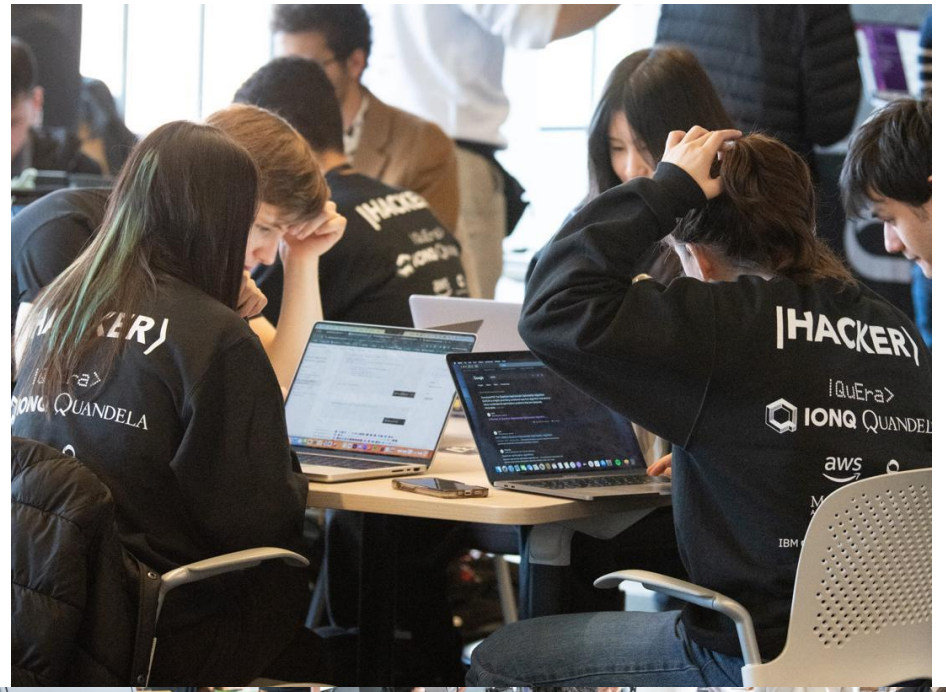
Next lectures will be for more advanced

What is a quantum hackathon?
A quantum hackathon is an event where participants use quantum computing tools and algorithms to tackle scientific, technological, or real-world challenges.

We will use IBM Qiskit to develop quantum programs, run them on simulators or quantum hardware, and investigate applications in:

- Chemistry and materials: Using the Variational Quantum Eigensolver (VQE) to calculate molecular energies.
- Healthcare and data science: Exploring quantum machine learning (QML) to classify data.
- Energy and electric grids: Investigating optimization problems using quantum or hybrid quantum-classical approaches.

During Hackathon week: you will receive a challenge, form teams, develop a solution, and present their results to judges. Projects are evaluated on criteria such as scientific understanding, creativity, implementation, teamwork, and presentation. Importantly, participants do not need to be quantum computing experts.

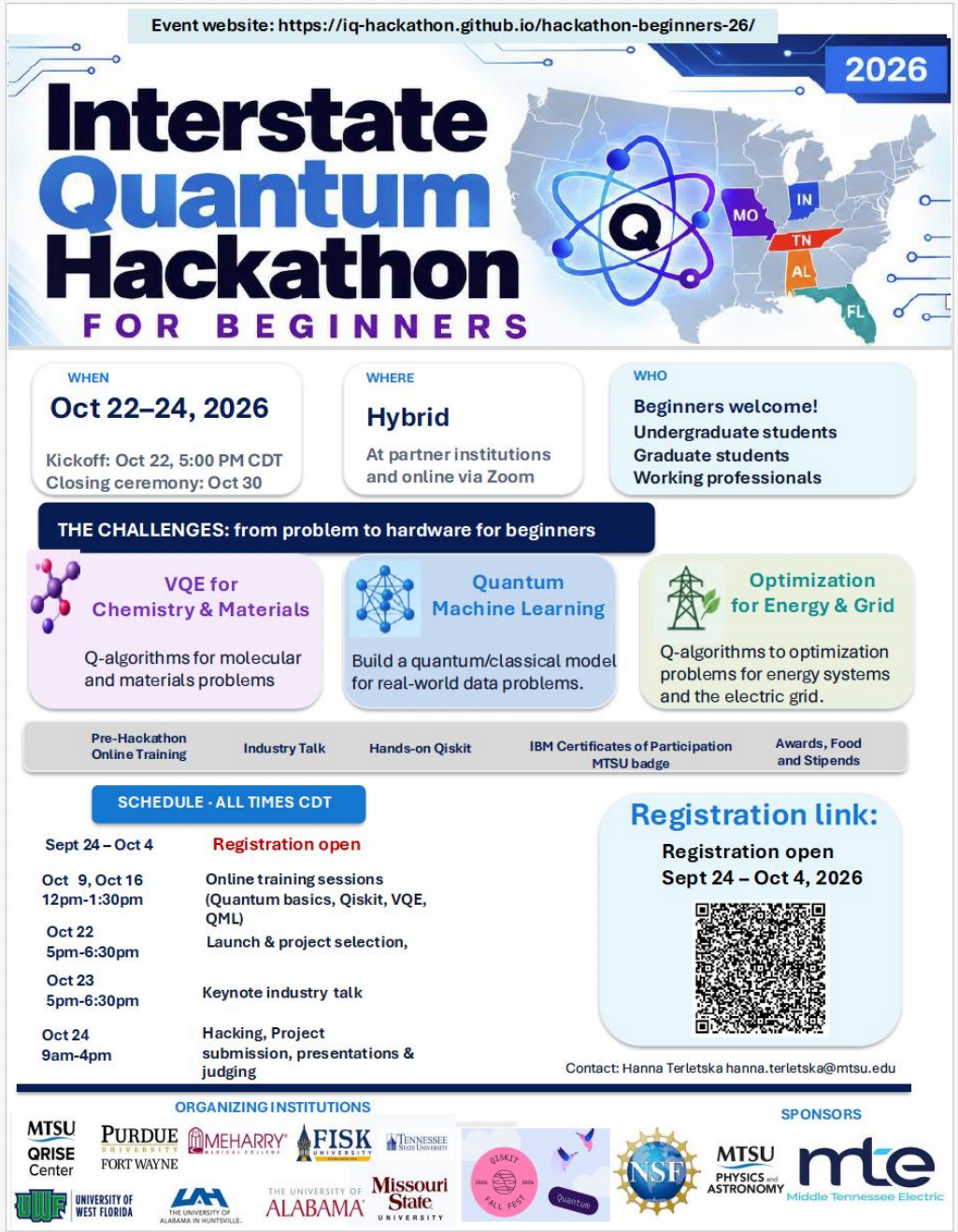




PART 1: Preparation (this and next week)

This week: Training 1: Quantum Foundations (Presenter Hanna Terletska)
Friday, October 9 | 12:00–1:30 PM CDT | Online (Zoom)
Our first session is specifically designed for **complete beginners and participants with very limited quantum background**. We will introduce the foundations you will need to get started, including **qubits, quantum states, superposition, quantum gates, circuits, and measurement**. The goal is to give you the conceptual foundation needed to begin working with quantum algorithms and Qiskit.
Zoom link:

Next week: Training 2: From Foundations to Quantum Algorithms (VQE, QML, QAOA)
Presenters (Abhijit Chakraborty, David Liu, Pushpita Chatterjee, Dinh Nguyen, Lei Qian, Varun Puram)
Friday, October 16 | 12:00–1:30 PM CDT | Online (Zoom)
Training 2 will move toward Qiskit, VQE, and Quantum Machine Learning, providing more direct preparation for the hackathon challenges.



PART2: Hackathon.

Then comes the Hackathon: October 22-24, 2026

1) **Thursday, October 22 at 5:00 PM CDT** – hackathon launch and project selection.

2) **Friday, October 23**- industry keynote

Zoom link:

3) **Saturday, October 24, from 9:00 AM–4:00 PM CDT (hybrid)** -the coding day, including project development, submission, presentations, and judging.

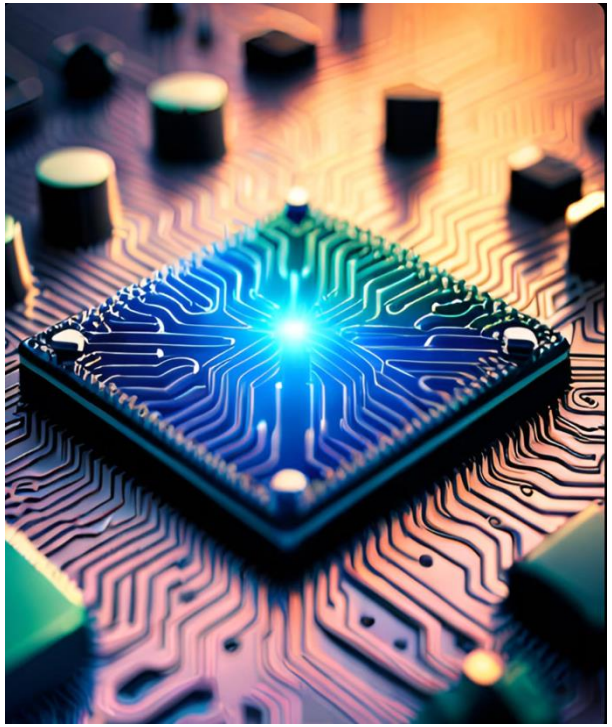
Zoom link:

You will choose from three challenge areas:

- **VQE for Chemistry & Materials**: quantum algorithms for molecular and materials problems
- **Quantum Machine Learning**: quantum/classical approaches to real-world data
- **Optimization for Energy & the Electric Grid**: quantum approaches to energy-system optimization

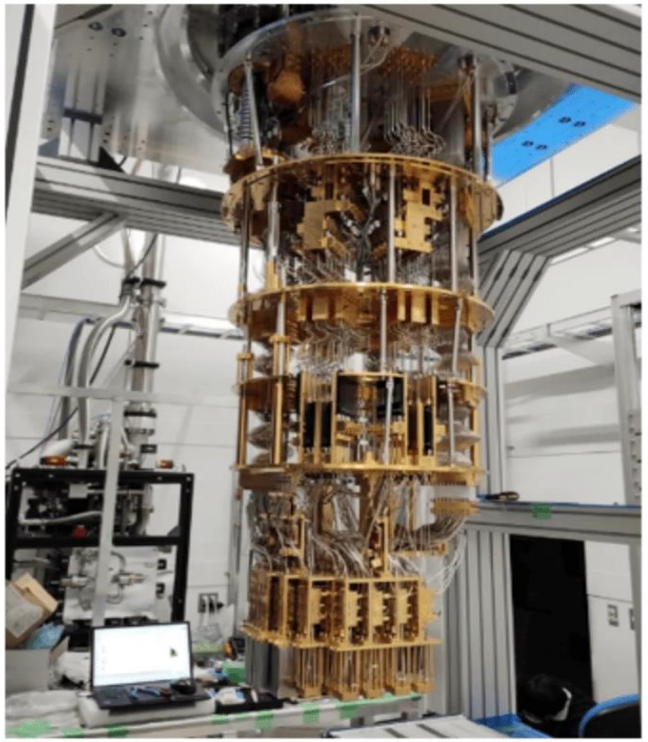


1. Why Quantum? Why Now?



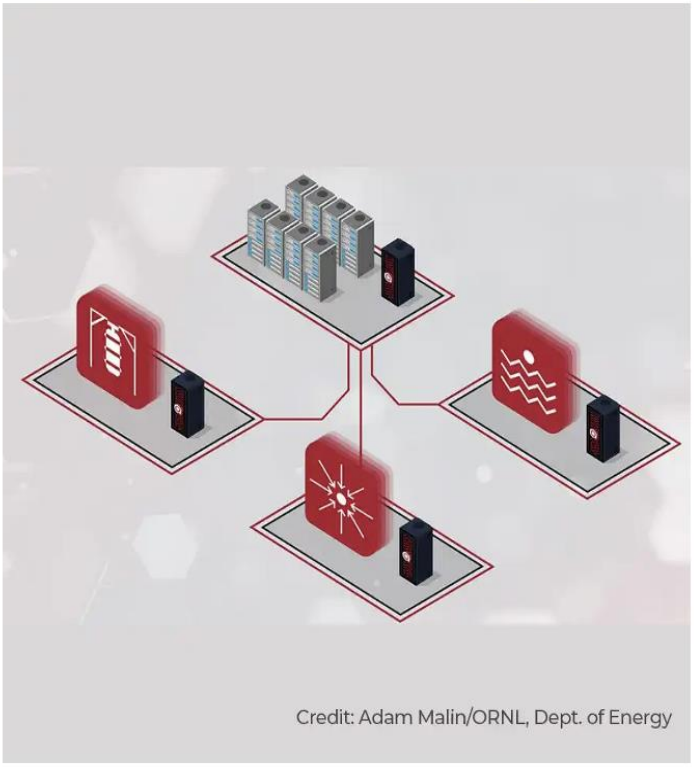
Quantum technology to solve hard problems

Q-Computers



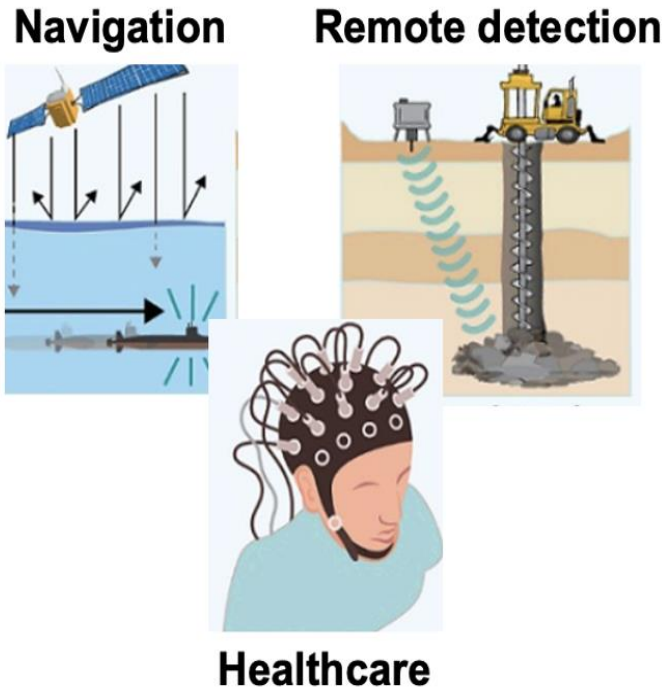
Efficiently solve problems which supercomputers cannot: *medicine, financial modeling, logistics, supply chain*

Q-Networking



Q-Communication:
Security and Defense

Q-Sensing



Ultra-sensitivity

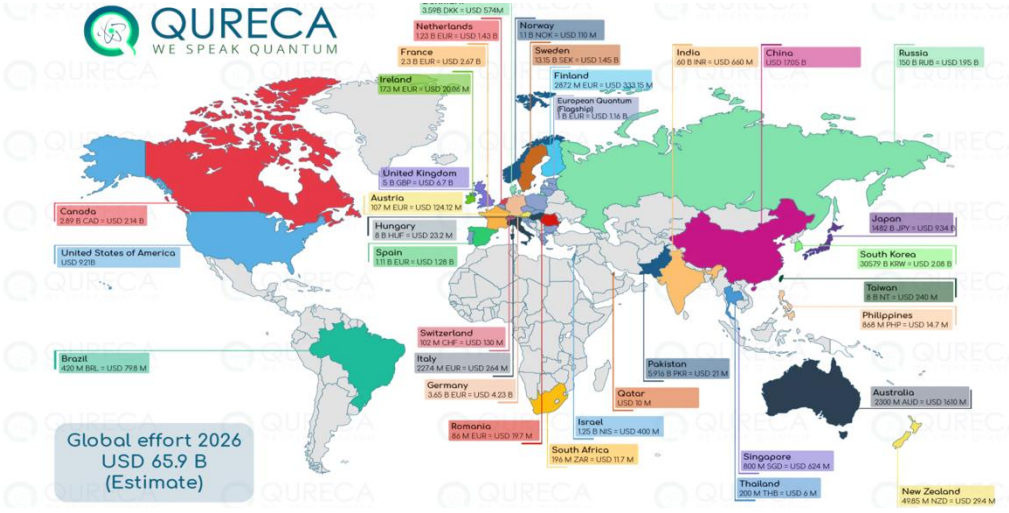
Developments in quantum physics are fundamentally changing the way we detect, transfer, and process information.

Global impact and race

McKinsey: Quantum computing could create up to $2.7 trillion of economic value worldwide by 2035.

Whoever leads in quantum will help shape the 21st century.

Quantum investments across the globe



CHINA
$15B
Government investment. World's largest quantum network (12,000 km). Two quantum satellites in orbit.

EUROPE
$10B
EU Quantum Flagship + national programs. 1/3 of all quantum companies worldwide. "Quantum Act" launching 2026.

UNITED STATES
$5B
Public investment. But leading in private sector: $1.7B in 2024 alone. 300 quantum startups vs China's 30.

Interdisciplinary

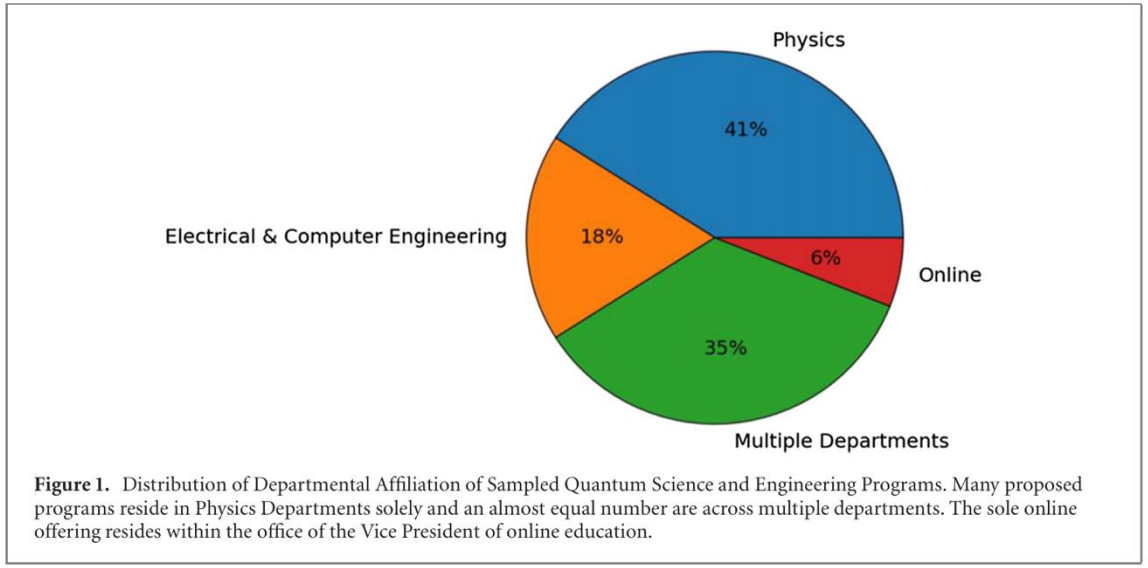


Figure 1. Distribution of Departmental Affiliation of Sampled Quantum Science and Engineering Programs. Many proposed programs reside in Physics Departments solely and an almost equal number are across multiple departments. The sole online offering resides within the office of the Vice President of online education.

- New jobs, new tech, economic development
- Big industries will benefit the most;
- Defense and nat. security matter

Quantum Industries







IONQ

D:wave

Quantum Computing

Users	Applications	Software offerings	QPUs²			Hardware
Material Science	Not categorized		Superconducting			Cryogenics
Finance			Ion Trap	Neutral Atoms		Lights and lasers
Life Sciences	Cloud access to QPUs	Simulators / q-inspired / etc	Silicon	Photonics		Other Componentry

Quantum industry:

1. **Hardware**
2. **Software**
3. **Applications and Users**
4. **Big industry and startups**

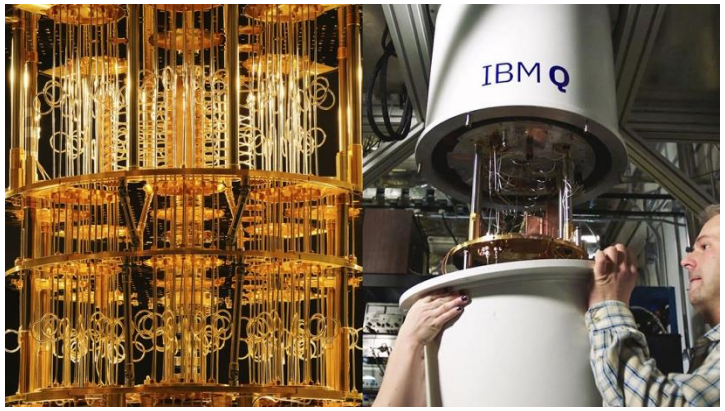
5. **Cases of early application of q-tech**

https://medium.com/dealflow-insights/the-rising-quantum-industry-current-state-and-future-prospects-fc8213922600

Why are we here today? To Jumpstart you in pathway in Quantum

Future Quantum Workforce is YOU

Work in Q-Industry



Work in Q-Laboratory





Q-Scientists



Quantum physicist Michelle Simmons

University Student



Today: fundamentals of QC

Q-Computing



See afternoon session for types of problems quantum could help with

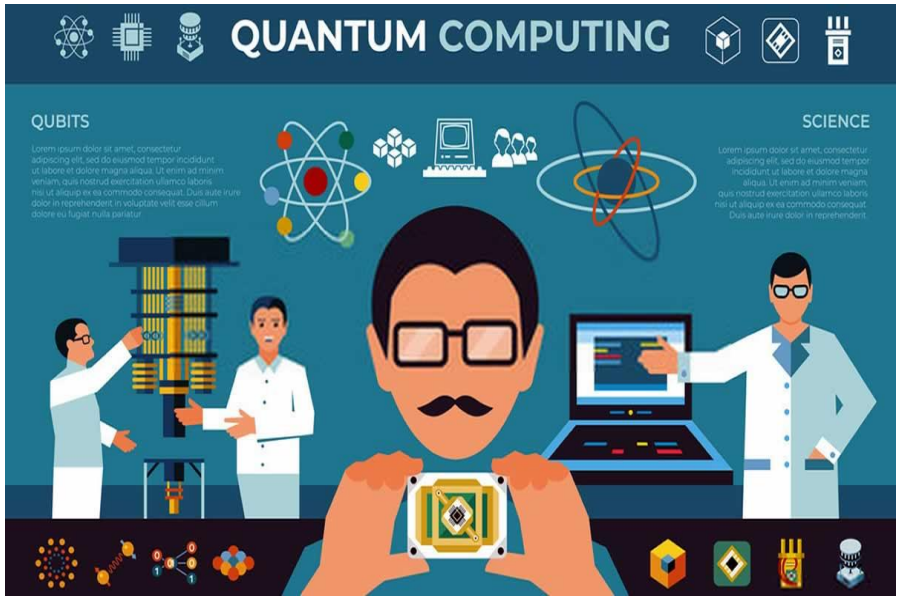
Q: Why do we need QC: Quantum computers will eventually...

A. Replace classical computers.

B. Increase the speed of conventional computers.

C. Classical Computer are bad, and we should get rid of them

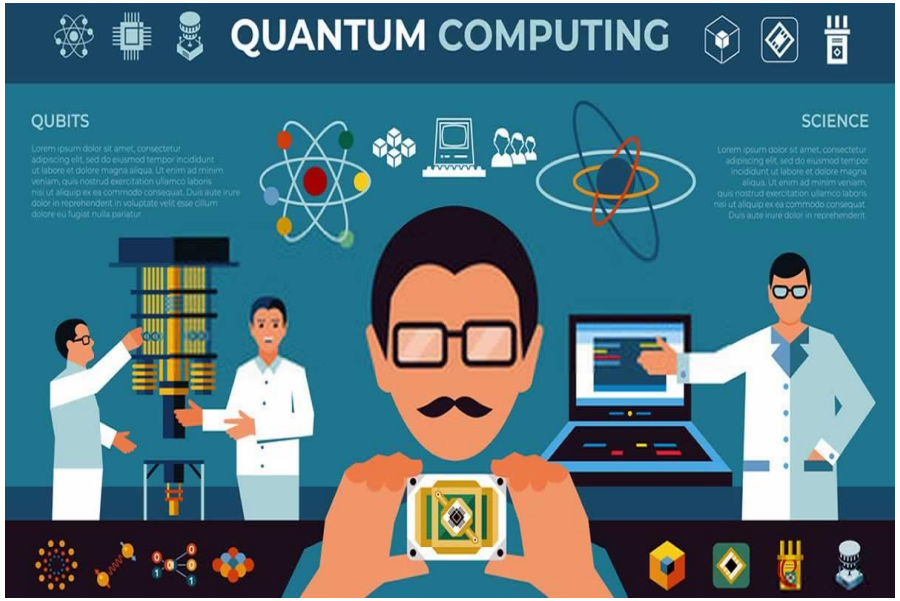
D. Solve problems that are too difficult for classical computers.



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Q: Why do we need QC: Quantum computers will eventually...

A. replace classical computers.
B. increase the speed of conventional computers.
C. Classical Computer are bad, and we should get rid of them
D. Solve problems that are too difficult for classical computers.



One of the biggest misconceptions is that quantum computers are simply better than CC. They are not.

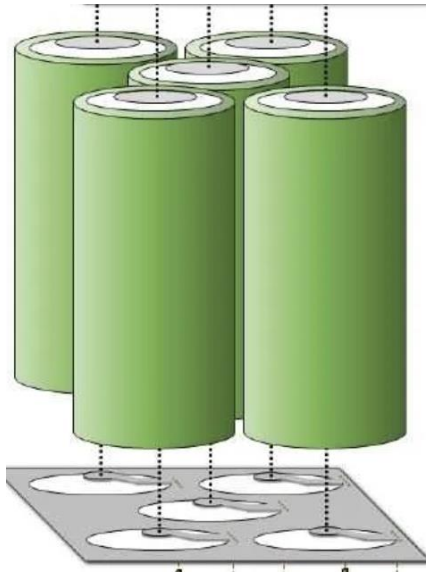
Not a faster laptops

Not replacing classical computers

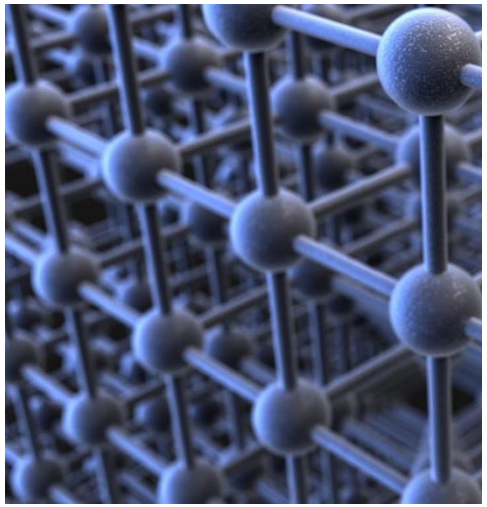
Not useful for everything

What type of problems QC might be useful for?

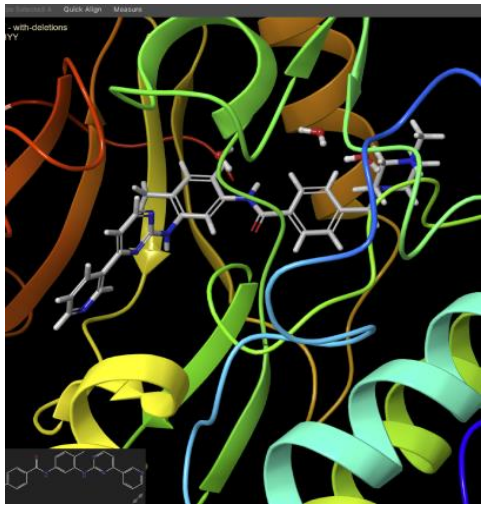
Today's supercomputers are incredibly powerful. Yet there remain **important problems that are extremely difficult**.



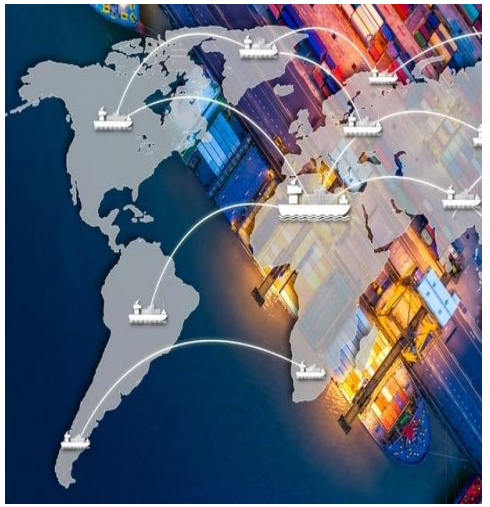
Battery



Materials



Drug molecules



Supply chain



Cybersecurity

Event website: https://iq-hackathon.github.io/hackathon-beginners-26/

Interstate Quantum Hackathon

2026

FOR BEGINNERS



WHEN

Oct 22–24, 2026

Kickoff: Oct 22, 5:00 PM CDT
Closing ceremony: Oct 30

WHERE

Hybrid

At partner institutions and online via Zoom

WHO

Beginners welcome!
Undergraduate students
Graduate students
Working professionals

THE CHALLENGES: from problem to hardware for beginners

VQE for Chemistry & Materials

Q-algorithms for molecular and materials problems

Quantum Machine Learning

Build a quantum/classical model for real-world data problems.

Optimization for Energy & Grid

Q-algorithms to optimization problems for energy systems and the electric grid.

Pre-Hackathon Online Training | Industry Talk | Hands-on Qiskit | IBM Certificates of Participation MTSU badge | Awards, Food and Stipends

SCHEDULE · ALL TIMES CDT

Sept 24 – Oct 4	Registration open
Oct 9, Oct 16 12pm-1:30pm	Online training sessions (Quantum basics, Qiskit, VQE, QML)
Oct 22 5pm-6:30pm	Launch & project selection,
Oct 23 5pm-6:30pm	Keynote industry talk
Oct 24 9am-4pm	Hacking, Project submission, presentations & judging

Registration link:

Registration open
Sept 24 – Oct 4, 2026

Contact: Hanna Terletska hanna.terletska@mtsu.edu

ORGANIZING INSTITUTIONS

MTSU QRISE Center | Purdue University Fort Wayne | Meharry Medical College | Fisk University | Tennessee State University | University of West Florida | The University of Alabama in Huntsville | The University of Alabama | Missouri State University

SPONSORS

NSF | MTSU Physics and Astronomy | mte Middle Tennessee Electric

Introduction

- Every computation has three elements: *data, operations, and results*



- In *Quantum Circuit*

Data ➔ Qubits

Operations ➔ Quantum Gates

Results ➔ Measurements

Data = Qubits

Classical Computer

- **Classical bit**



0 or **1**

can be in **one states at a time**

Classical variable:
Group of n bits stores **only one out of 2^n** possible values

Quantum Computer

- **Quantum bit=Qubit**

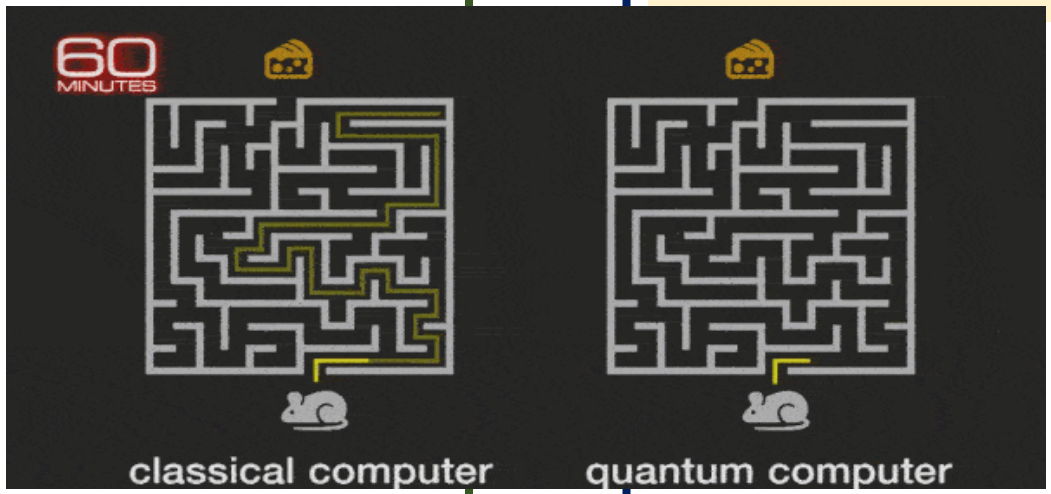


|0> and |1>

can be in a **superposition of two states** **|q>=a|0>+ b|1>**

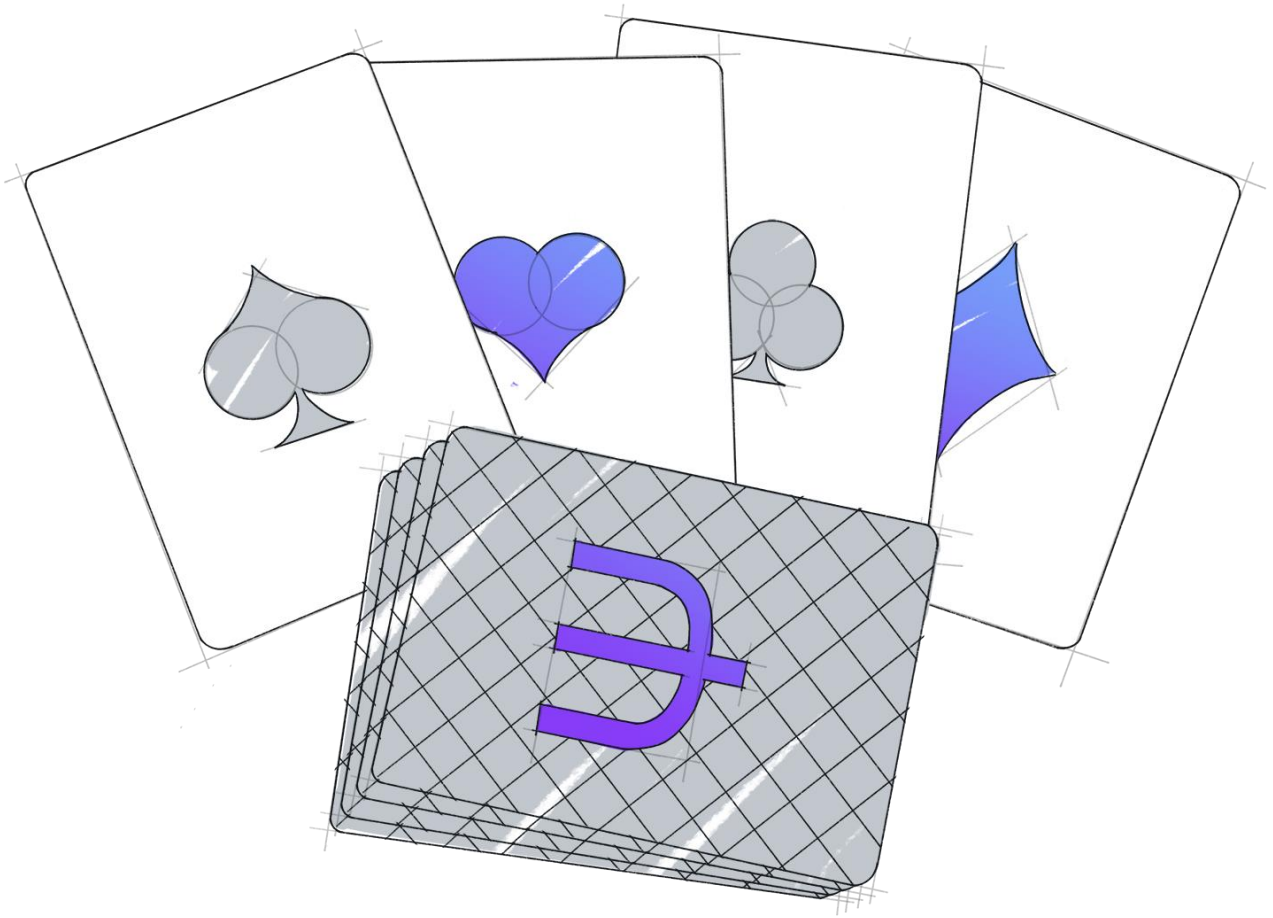
- **Quantum variable:**

Group of n qubits stores **all 2^n possible values simultaneously**



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Probability is natural property of a quantum system.



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Qubit: Dirac notation- bra-ket notation



Paul Dirac

$$|\psi>$$

Qubit=Ket psi

$$|\psi>= a|0> +b|1>$$

a Ket 0 + b ket 1

ket

Probability amplitude

a-probability amplitude for |0>

$|a|^2$ **-probability of measuring |0>**

b-probability amplitude for |1>

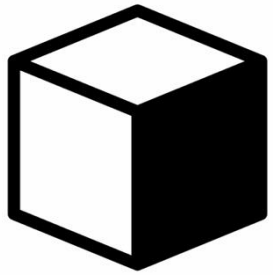
$|b|^2$ **-probability of measuring |1>**

$$|a|^2 + |b|^2 = 1$$

- **Probability adds up to 1 (100%)**

Note, a bra is opposite to ket $< q|$

Qubit: bra-ket notation (math)



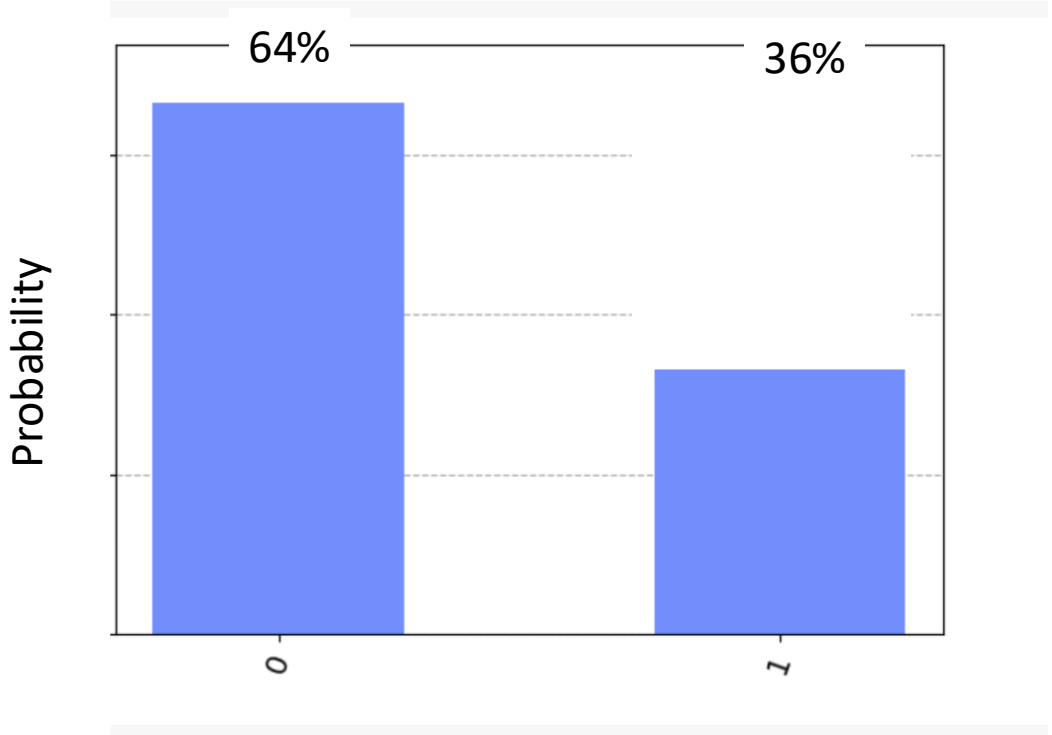
$$|\psi> = 0.8|0> + 0.6|1>$$

$$|\psi> = \alpha|0> + \beta|1>$$

α-probability amplitude for |0>

β- probability amplitude for |1>

$$|\alpha|^2 + |\beta|^2 = 1$$



$$|\psi> = a|0> + b|1>$$

Q: in this qubit state

$$\left|\psi> = \frac{1}{2}\right| 0> + \frac{\sqrt{3}}{2}|1>$$

What is the probability that after measurement you will get a |0> state?

A) ½

B) ¾

C) ¼

D) $\frac{\sqrt{3}}{2}$

Probabilistic nature of a qubit

$$|\psi> = a|0> + b|1>$$

Q: in this qubit state

$$|\psi> = \frac{1}{2}|0> + \frac{\sqrt{3}}{2}|1>$$

What is the probability that after measurement you will get a |0> state?

A) ½

B) ¾

C) ¼

D) $\frac{\sqrt{3}}{2}$

Qubit as a vector in a Hilbert Space; Basis vectors

- In a vector form:

Vector is written as a column matrix

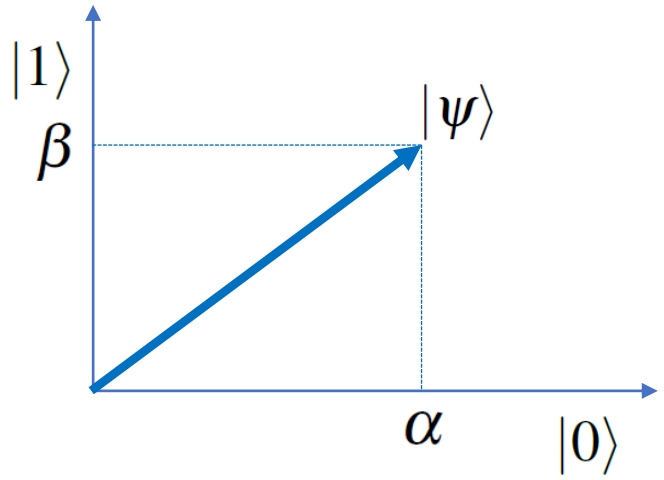
$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Computational (Z) basis

$$\mathscr{S} = \{|0\rangle, |1\rangle\}$$

- *Superposition* between 0 and 1

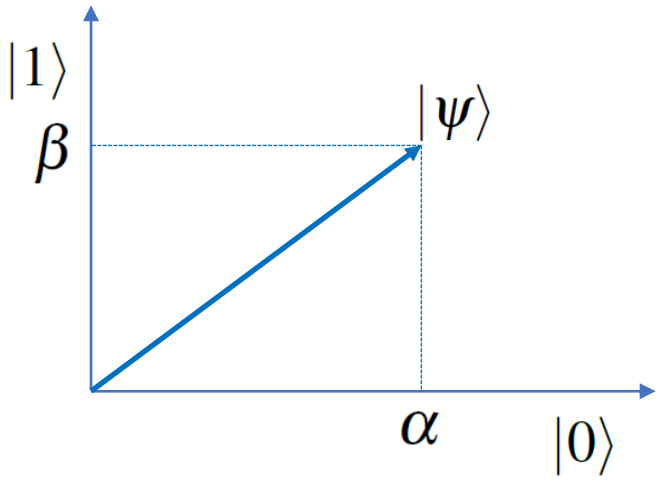
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$



$|\alpha|^2 + |\beta|^2 = 1$ $|\psi\rangle$ is normalized

Basis Vectors

- Quantum bits = Qubits
  - $|0\rangle$ and $|1\rangle$ are *orthonormal*
  - $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$



Their inner product is zero

bra-ket → $\langle b|a\rangle = b_1^* a_1 + b_2^* a_2$

$\langle 0|1\rangle = \begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} =$

$1x0 + 0x1 = 0 = \langle 1|0\rangle$

Orthonormal basis

$\langle 0|0\rangle = \langle 1|1\rangle = 1$

Basis Vectors in Hadamard Basis

- Hadamard Basis:

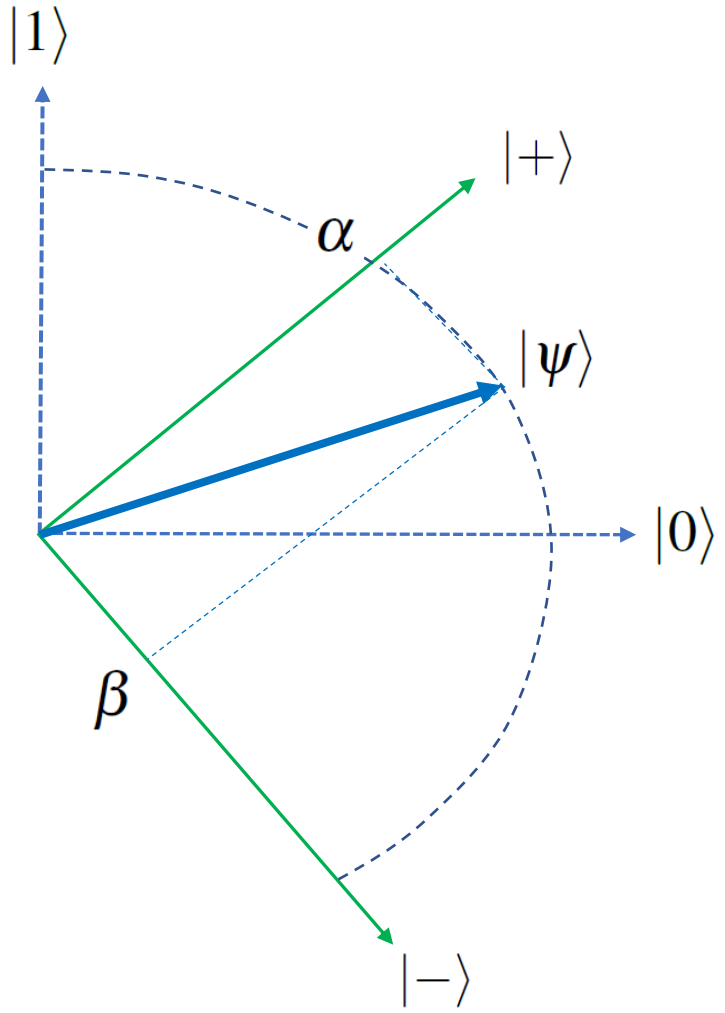
$$\mathcal{H} = \{|+\rangle, |-\rangle\}$$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

superposition

$$|\psi\rangle = \alpha|+\rangle + \beta|-\rangle$$



Note $\langle +|-\rangle = 0$ $\langle +|+\rangle = \langle -|-\rangle = 1$

See Notebook 2 for ,more exercises

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

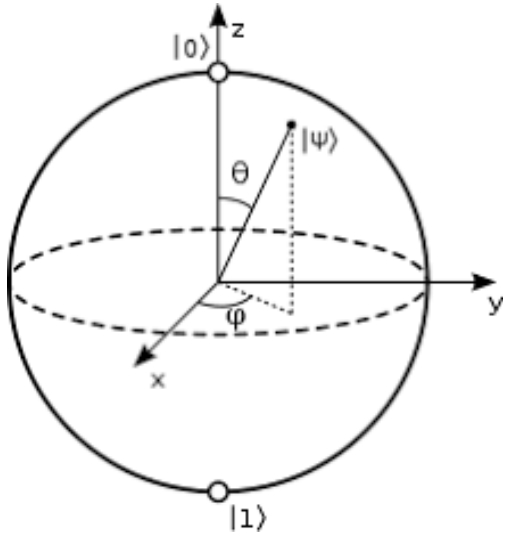
3D Representation-Bloch sphere

$$|\psi >= a|0 > +b|1 >$$

To account for all possible states in 3D, we can represent qubit as

$$|\psi >= \cos\frac{\theta}{2}|0 > +\sin\frac{\theta}{2}e^{i\phi}|1 >$$

Relative Phase



-theta measured from z-axis
-phi-in x-y plane

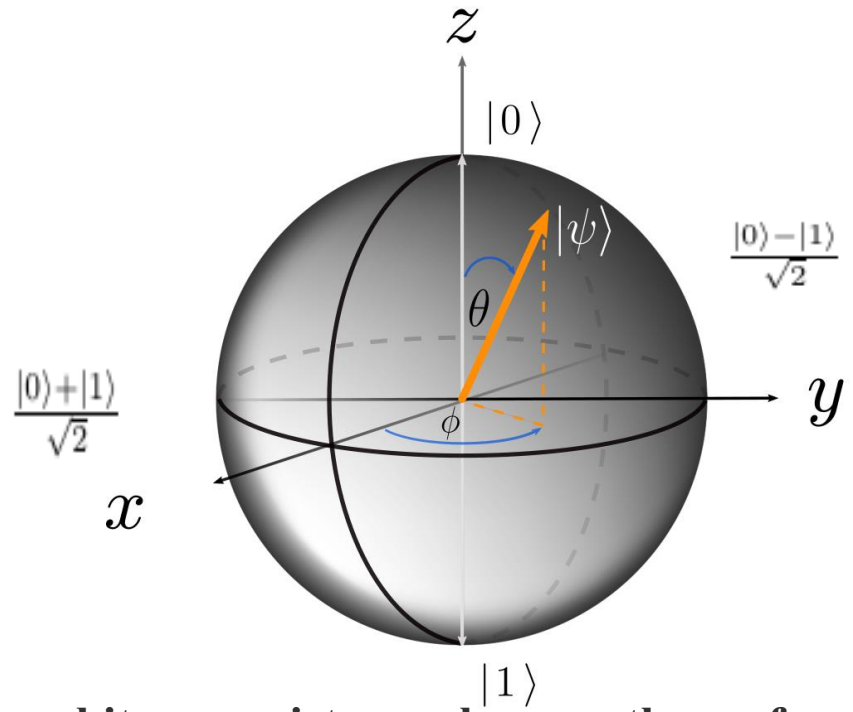
Bit



By convention, the “north pole" of the sphere represents state 0 and the “south pole" represents state 1.
A classical bit is either at the north pole or the south pole, but nowhere else.

Qubit

$$|\psi>=\cos\frac{\theta}{2}|0>+\sin\frac{\theta}{2}e^{i\phi}|1>$$



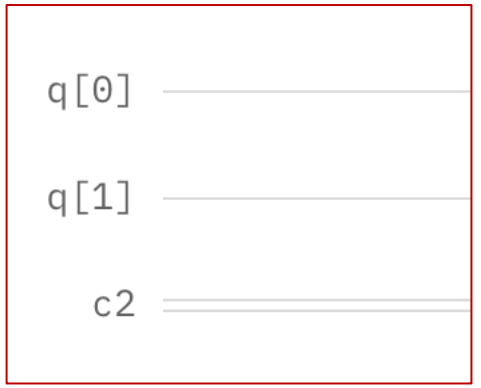
A qubit may exist anywhere on the surface of the sphere.
When the qubit state is anywhere *except* for the north and south poles, it is in a **superposition** state.

Copyright: Hanna Terletska, MTSU.

2-Qubit States

$|a\rangle$—
$|b\rangle$—

$|\psi\rangle_{ba}$ = |b> ⊗ |a> = |ba>



|0> —
|0> —
|00>

|1> —
|0> —
|01>

|0> —
|1> —
|10>

|1> —
|1> —
|11>

General two-qubit state

Superposition

$$|\Psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

Probability amplitude when we measure both qubits

$$\sum_{ij} |\alpha_{ij}|^2 = |\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$$

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Measuring probability of 2-qubit states

$$|\Psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

$$\sum_{ij} |\alpha_{ij}|^2 = |\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$$

If we measure two qubits, we will get:

- |00> with probability $|\alpha_{00}|^2$
- |01> with probability $|\alpha_{01}|^2$
- |10> with probability $|\alpha_{10}|^2$
- |11> with probability $|\alpha_{11}|^2$

Separable (Product) vs Entangled states

$$|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\psi\rangle_B \quad - \quad \text{product state,}$$

$$|\psi\rangle_{AB} \neq |\psi\rangle_A \otimes |\psi\rangle_B \quad - \quad \text{entangled state.}$$

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Combining two single-qubit systems

Separable states

Example

We have two single qubit states

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|\phi\rangle = |0\rangle$$

A two-qubit state is a tensor product of these vectors $|\psi\rangle \otimes |\phi\rangle$

Show that these states are separable (can be presented as a product)

Combining two single-qubit systems

Separable states

Example

We have two single qubit states

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|\phi\rangle = |0\rangle$$

A two-qubit state is a tensor product of these vectors

$$|\psi\rangle \otimes |\phi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

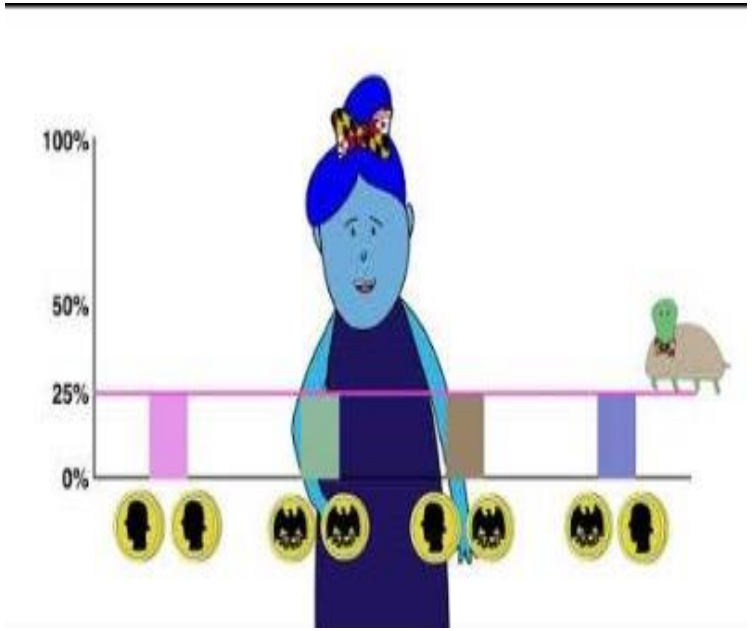
Show that these states are separable (can be presented as a product)

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Not all states can be separable-Entangled states

Entangled states- **collective identity** that can persist even if they're **separated by vast distances.**

Entangled objects: when one member of an entangled collection is measured, its answer is always tightly linke to the answers of the other objects in the group. Scientists call the connection shared by entangled particles "entanglement."



$$|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\psi\rangle_B \quad - \quad \text{product state,}$$
$$|\psi\rangle_{AB} \neq |\psi\rangle_A \otimes |\psi\rangle_B \quad - \quad \text{entangled state.}$$

Bell states or EPR pairs

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$
$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$
$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

All Bell states are maximally entangled!

Copyright: Hanna Terletska, MTSU.

Quantum Computing: Elements of Quantum Circuit

- **Every computation has three elements: *data, operations, and results***



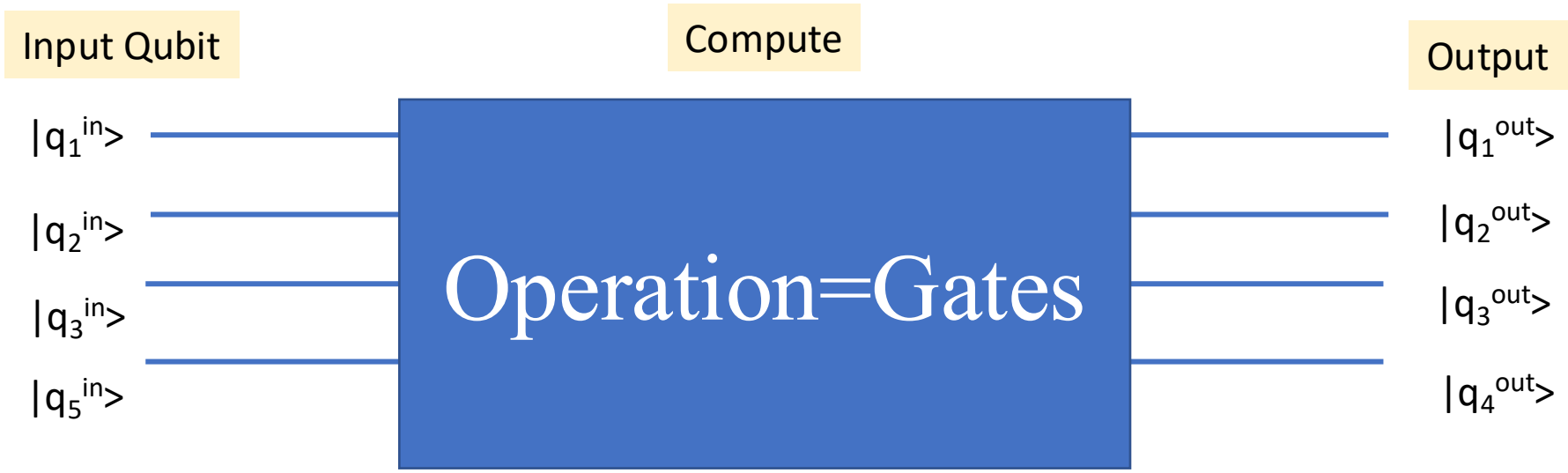
- **In *Quantum Circuit***

Data ➔ Qubits

Operations ➔ Quantum Gates

Results ➔ Measurements

Acting on a qubit

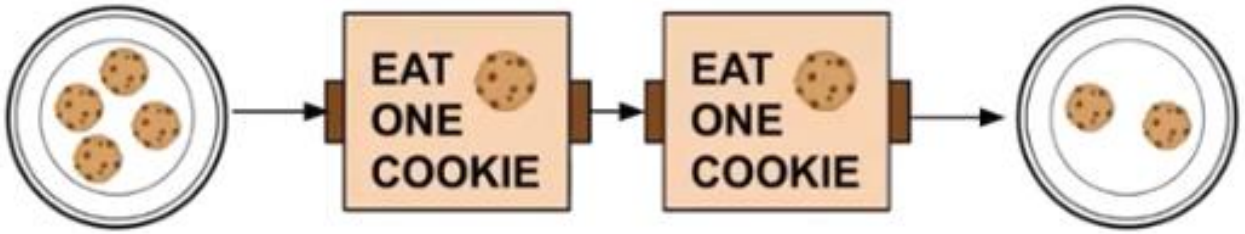


Quantum Computing

Operations = Quantum Gates

Operation: set of instructions





Operation order: from left to right

Slide borrowed From Diana Franklin

Single Qubit Gates



Gate Representation:



$$U|\psi_{in}> = |\psi_{out}>$$

U is a Unitary Matrix

$$U^\dagger U = U U^\dagger = \mathbb{I}$$

Complex conj. + transpose $U^\dagger = U^{-1}$

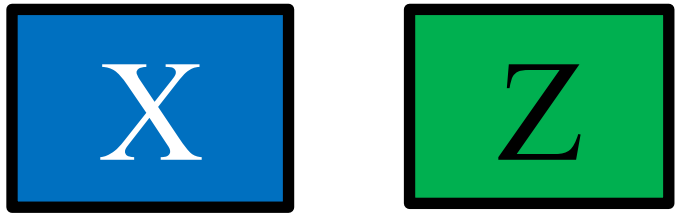
Unitary operators also preserve the norm of vectors. The norm of a vector remains unchanged when operated upon by a unitary operator.

Quantum gates are reversible $UU^{-1} = I$

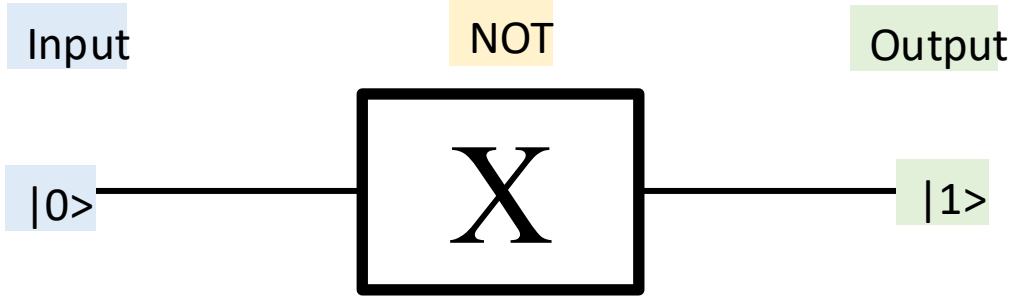
Quantum Algorithm = sequence of *quantum gates*

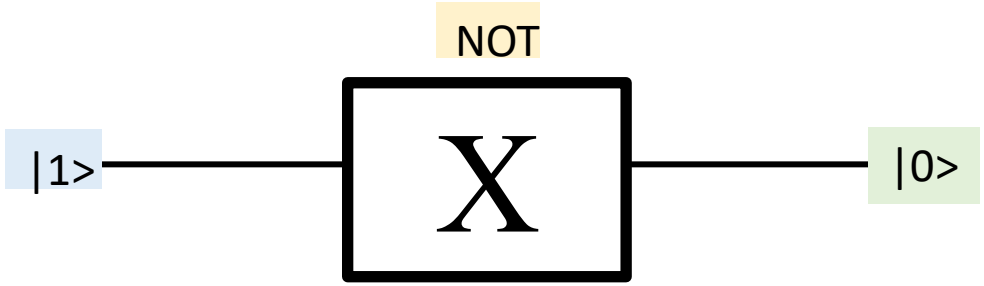
I. Single Qubit Gates

II. Multiple Qubit Gates

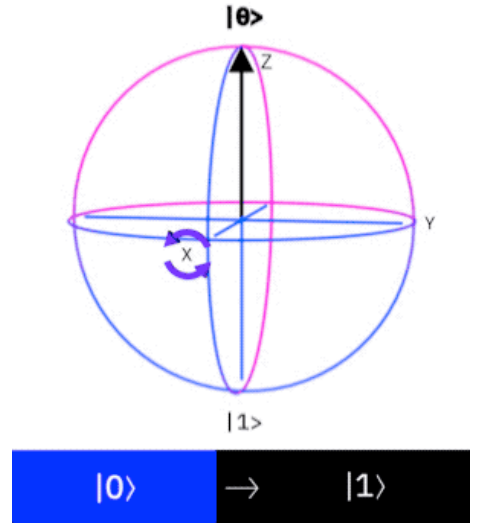


Quantum NOT gate (**qubit flip** gate)





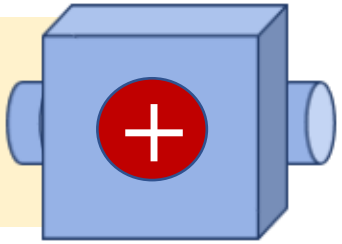
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

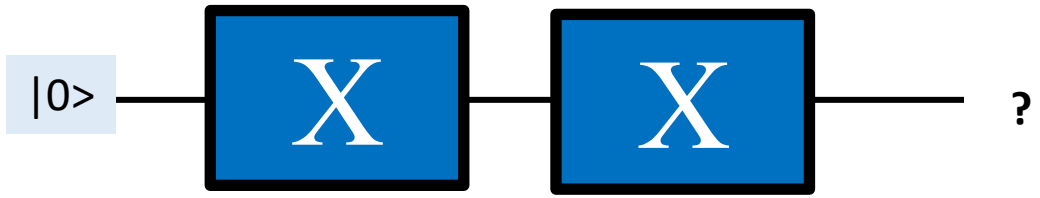


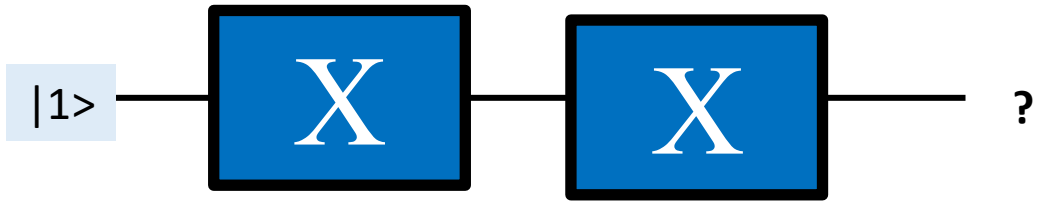
$$X|0\rangle = |1\rangle$$
$$X|1\rangle = |0\rangle$$

turns the $|0\rangle$ state into the $|1\rangle$ state, and the $|1\rangle$ state into the $|0\rangle$ state

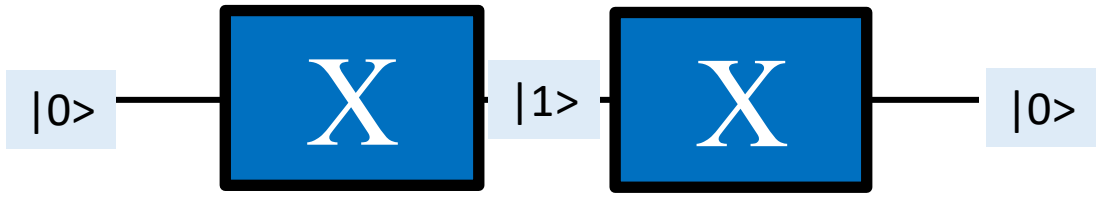
What would be the outcomes of the following operations?

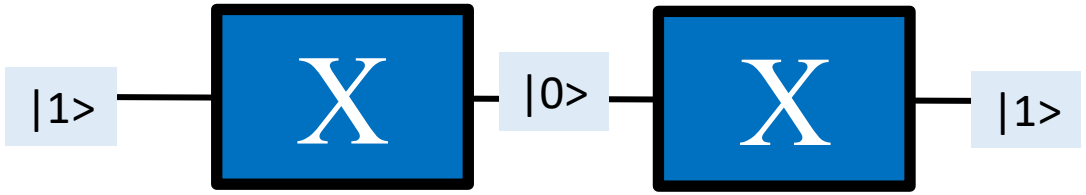






What would be the outcomes of the following operations?



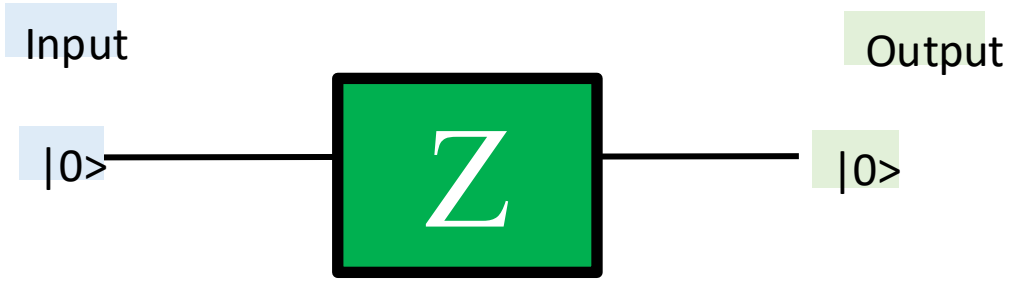


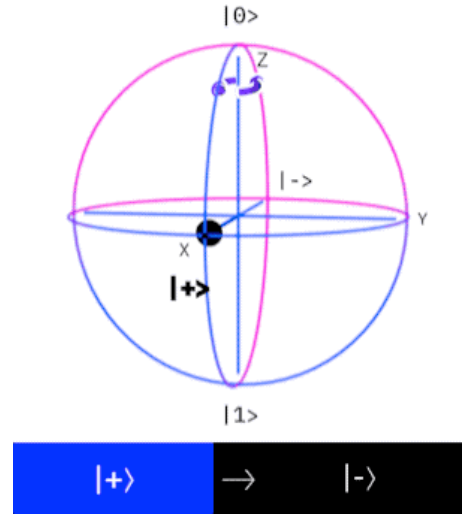
Two times of X cancel out→
brings you back to the original state

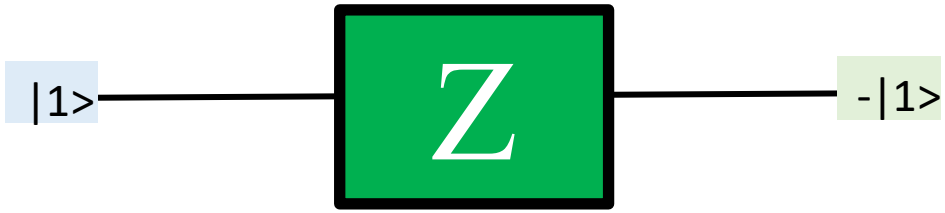
X is a reversible gate

Quantum Z gate (phase flip gate)

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$







$Z|0\rangle = |0\rangle$

$Z|1\rangle = -|1\rangle$

Flips phase of |1> only!

Several gates operation

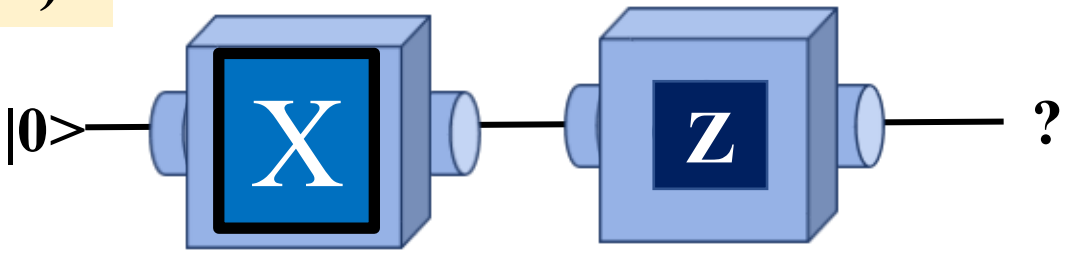


Several gates operation

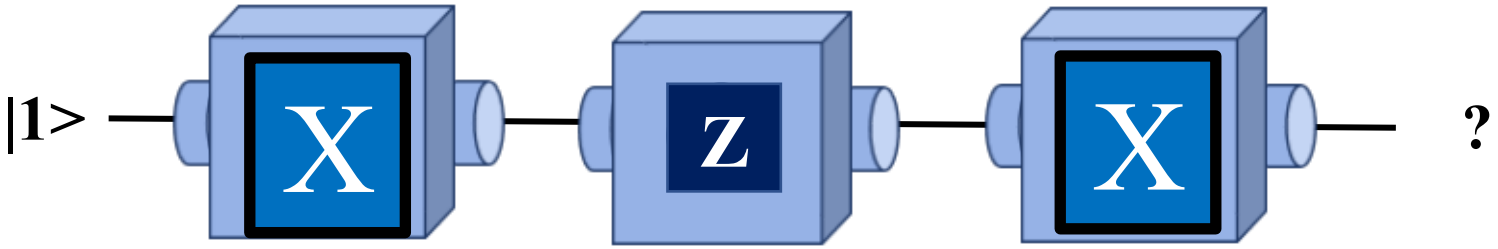


Task: Find the output for the circuit

5)

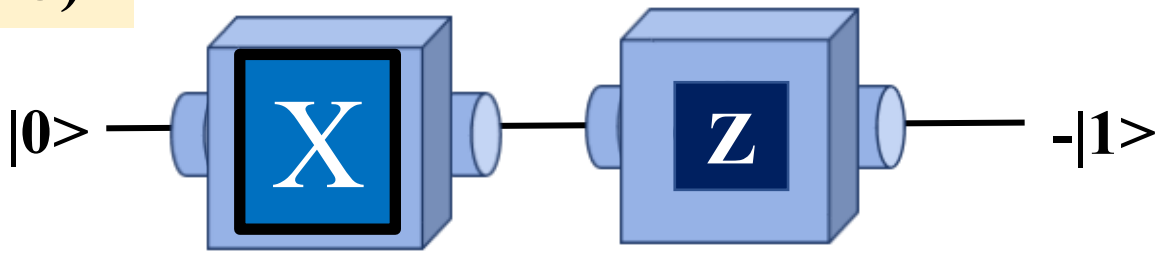


6)

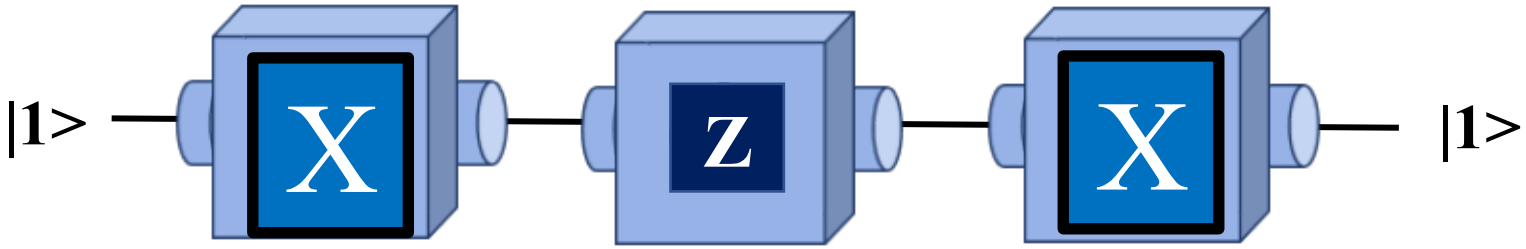


Task: Find the output for the circuit

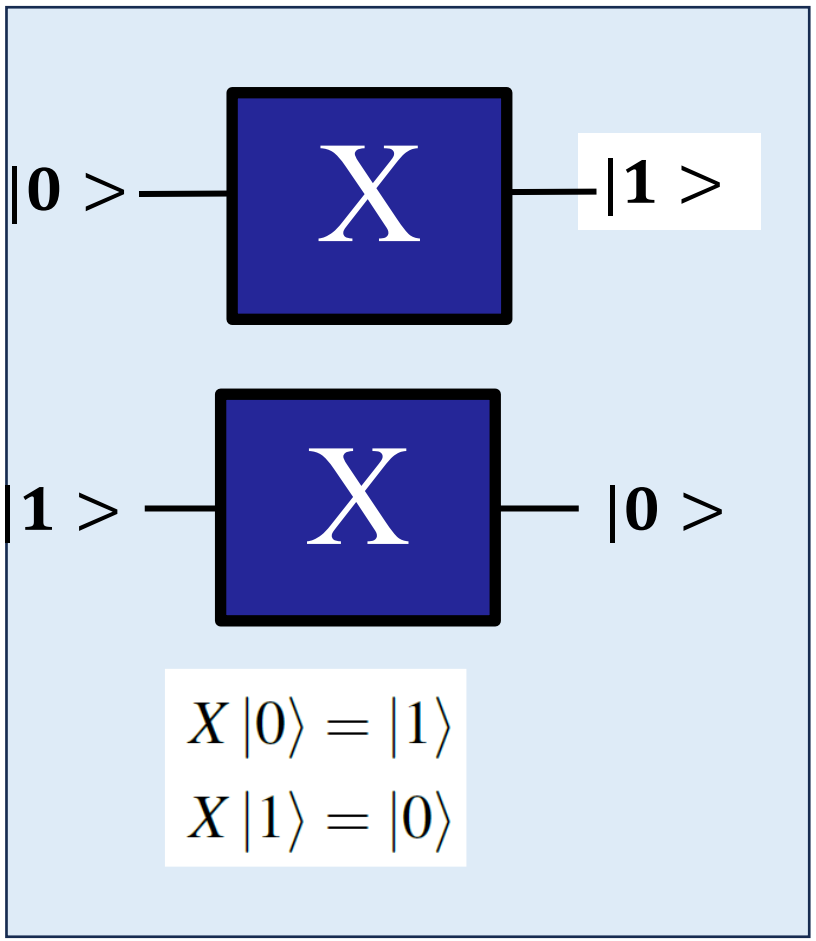
5)



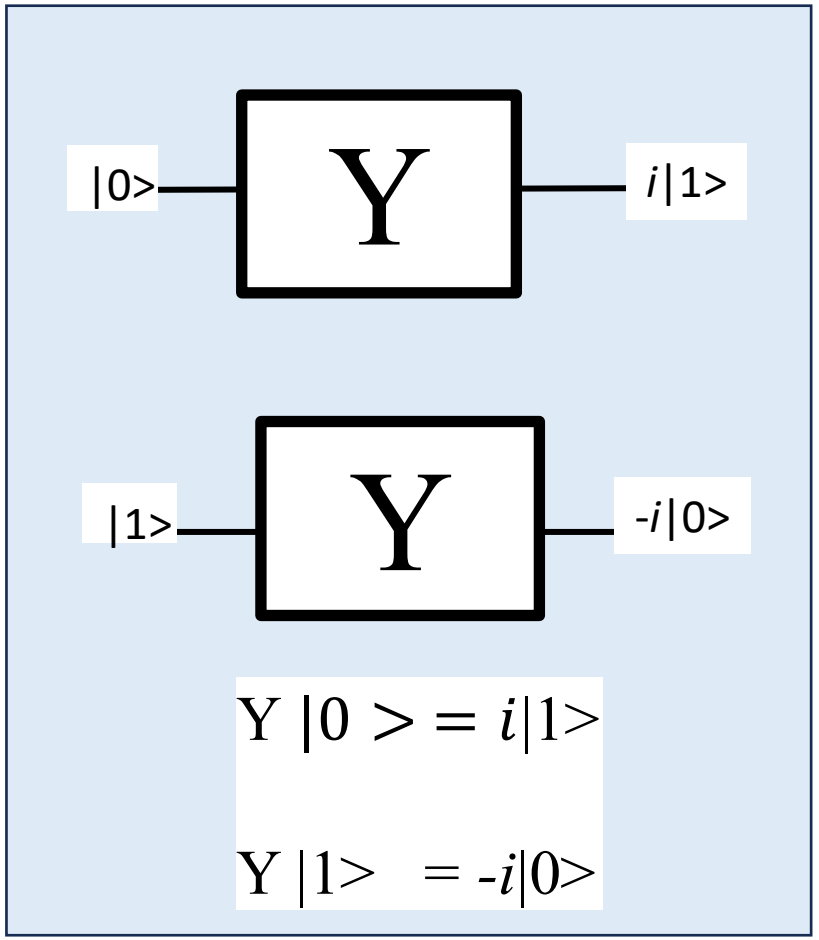
6)



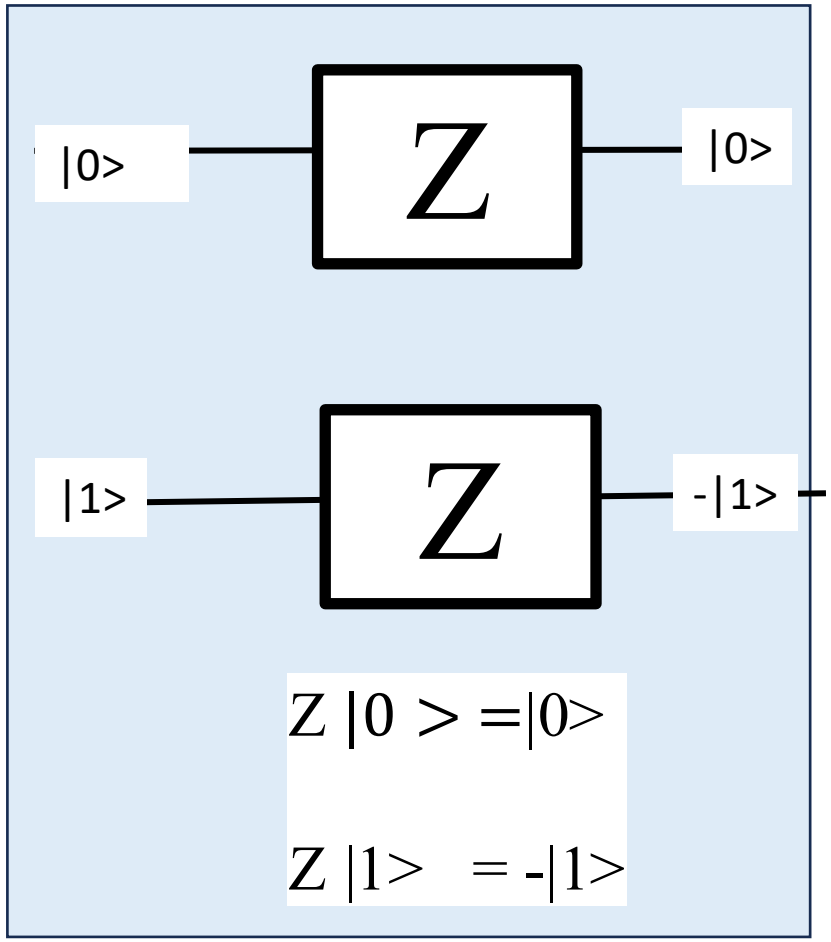
Single qubit gates



$$X|0\rangle = |1\rangle$$
$$X|1\rangle = |0\rangle$$



$$Y|0> = i|1>$$
$$Y|1> = -i|0>$$



$$Z|0> = |0>$$
$$Z|1> = -|1>$$

Rotation gates

The Rx gate $R_x(\theta) = e^{-i\frac{\theta}{2}X}$
is a single-qubit rotation through angle theta (radians) around the x-axis (clockwise).

$$R_x(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

The Ry gate $R_y(\theta) = e^{-i\frac{\theta}{2}Y}$
is a single-qubit rotation through angle theta (radians) around the y-axis.

$$R_y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

The Rz gate $R_z(\theta) = e^{-i\frac{\theta}{2}Z}$
is a single-qubit rotation through angle theta (radians) around the z-axis.

$$R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$$

Useful formula:

$$e^{-i\frac{\theta}{2}X} = \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right) X$$

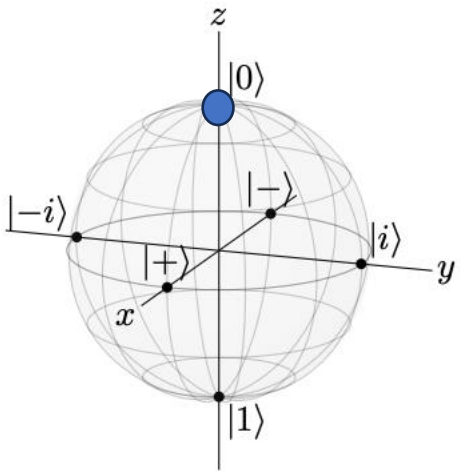
Hadamard gate:
puts a qubit into a superposition state

H-gate: superposition

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

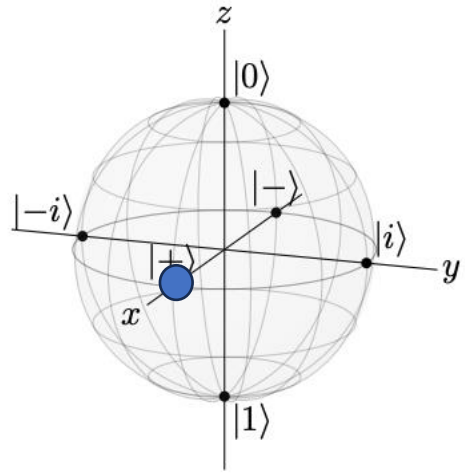
|0> — H — $\frac{1}{\sqrt{2}}$(|0>+|1>)

|1> — H — $\frac{1}{\sqrt{2}}$(|0>-|1>)



|0>

H



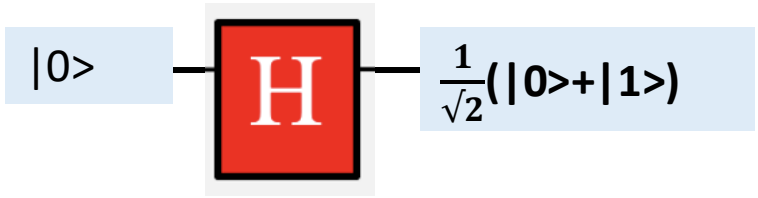
$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

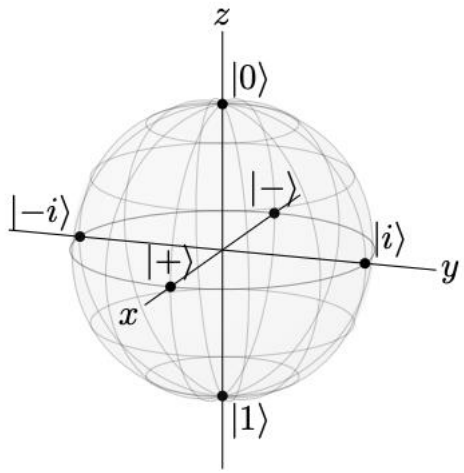
H |0 > =|+>

H |1> = |->

H-gate

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$



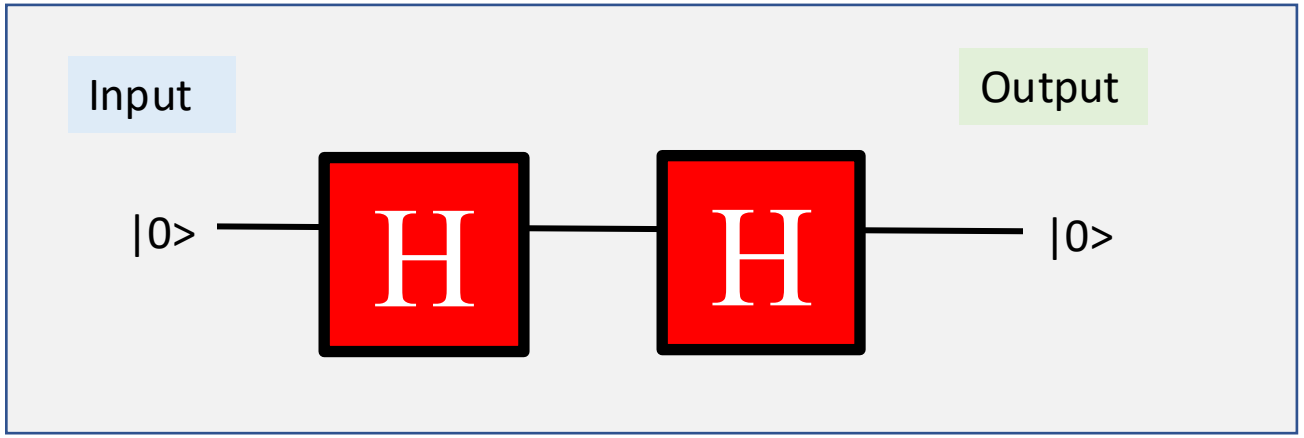


$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}|0> = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

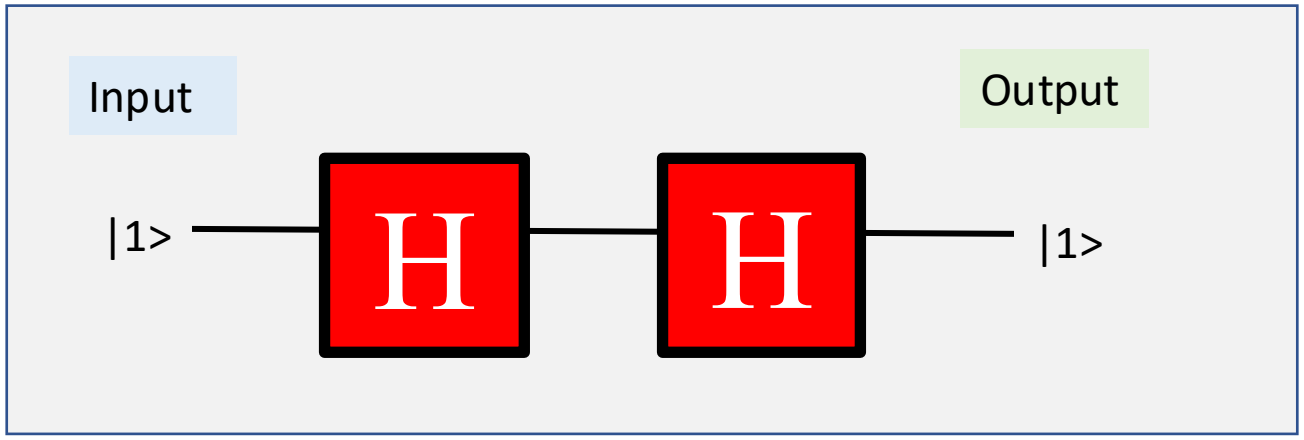
$$= \frac{1}{\sqrt{2}}\begin{pmatrix} 1x1 + 1x0 \\ 1x1 - 1x0 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0>+|1>) = |+>$$

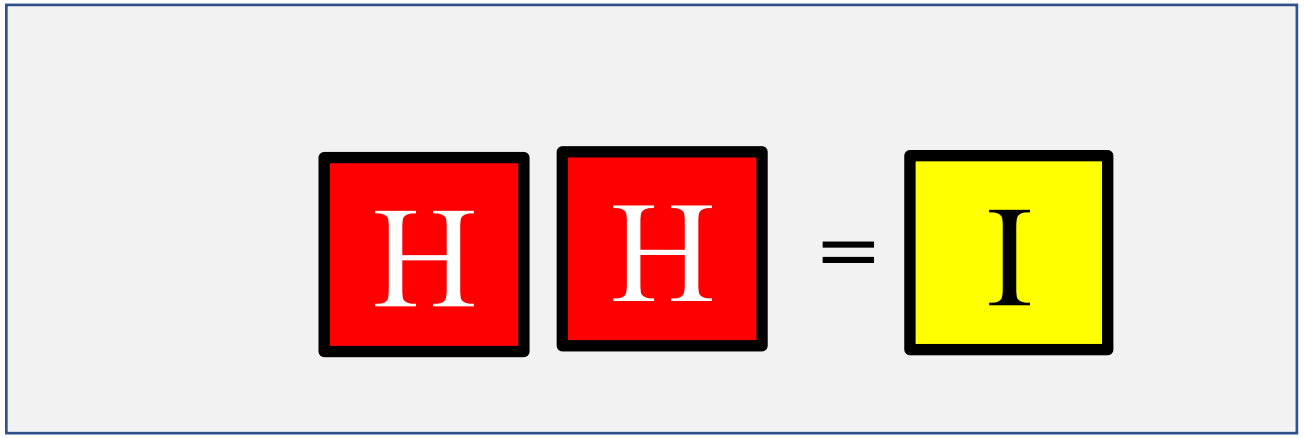
Two Quantum H gates: cancel superposition: HH=I



$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$



Let’s use math representation to show that



$$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}=$$

$$\frac{1}{2}\begin{pmatrix}1x1+1x1 & 1x1-1x1\\ 1x1-1x1 & 1x1+1x1\end{pmatrix}$$

$$=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=\mathbf{I}$$

Identity does not change anything

Let’s practice what we have learned today

Two qubit gates

CNOT Operation

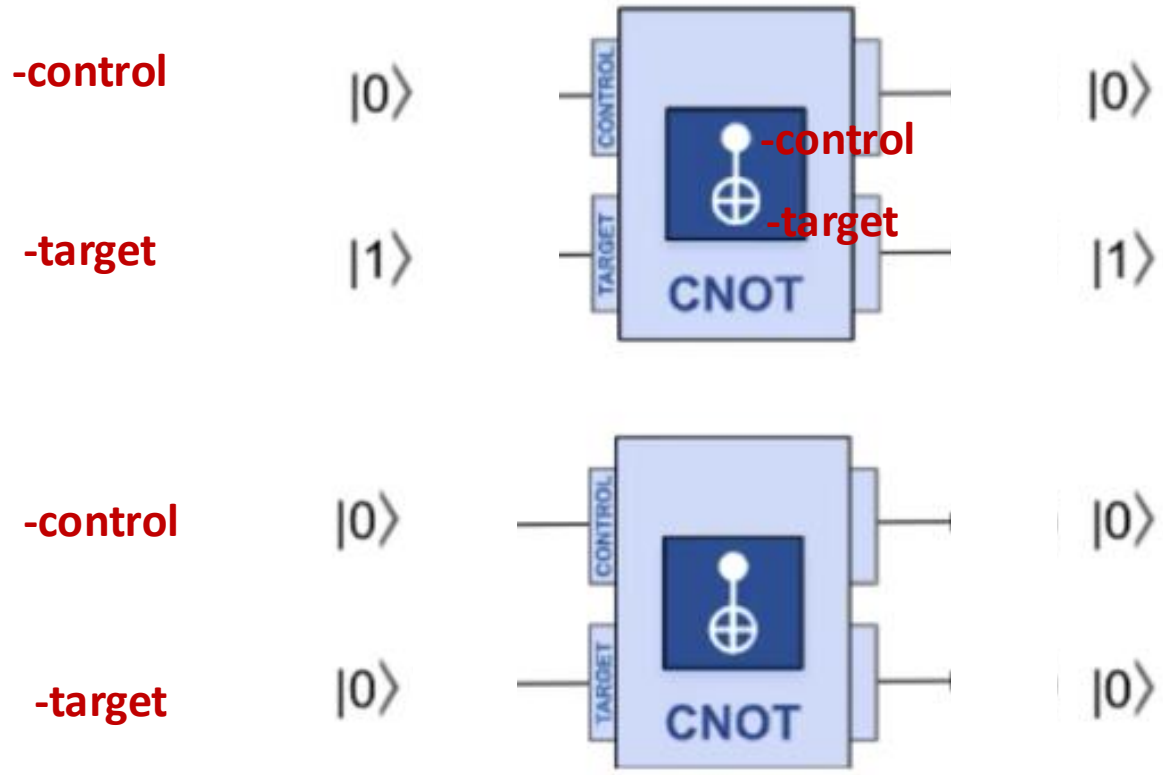
We have two inputs. One is called the **target qubit** and one is called the **control qubit**.

Control qubit determines if the target qubit changes



CNOT (controlled NOT)

CNOT Operation



Rules:

If the control qubit is |0> then the target does not change.

Control qubit itself does not change

|ct> |ct>

$$|00\rangle \rightarrow |00\rangle$$

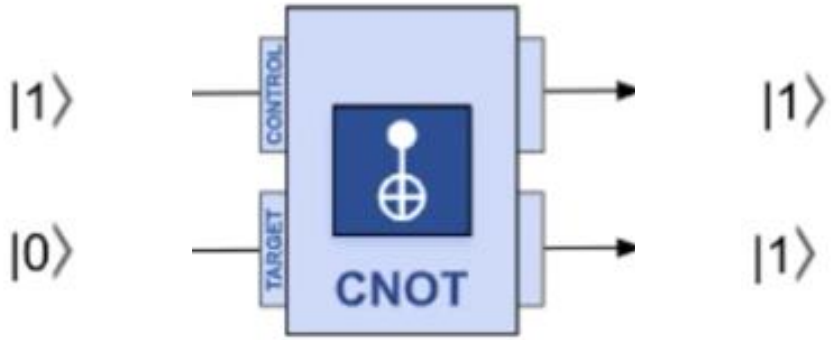
$$|01\rangle \rightarrow |01\rangle$$

CNOT Operation

Control itself does not change

-control

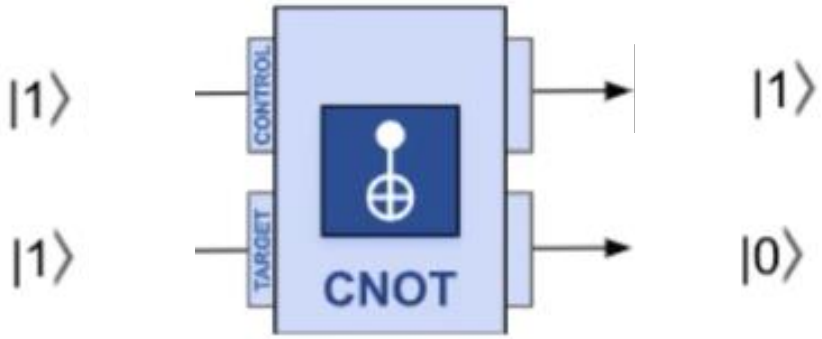
-target



If the control is |1> then the target qubit changes (a NOT gate is applied to the target gate).

-control

-target



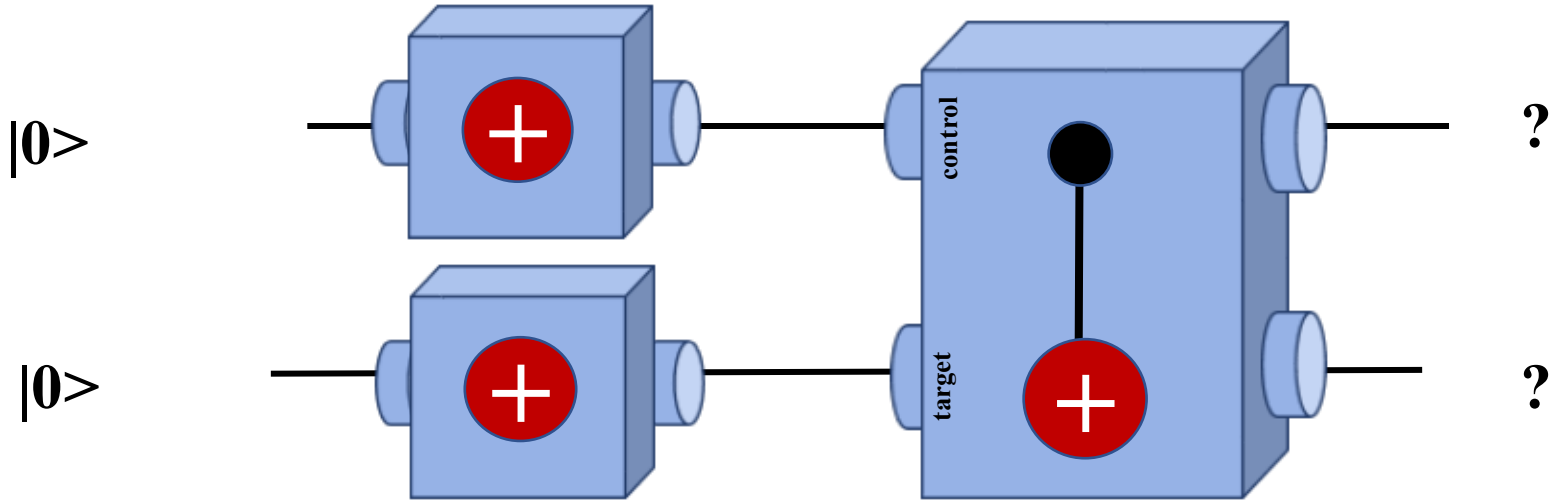
CNOT|ct> |ct>

$$|00\rangle \rightarrow |00\rangle$$
$$|01\rangle \rightarrow |01\rangle$$
$$|10\rangle \rightarrow |11\rangle$$
$$|11\rangle \rightarrow |10\rangle$$

3)

Task: Find the output for the circuit

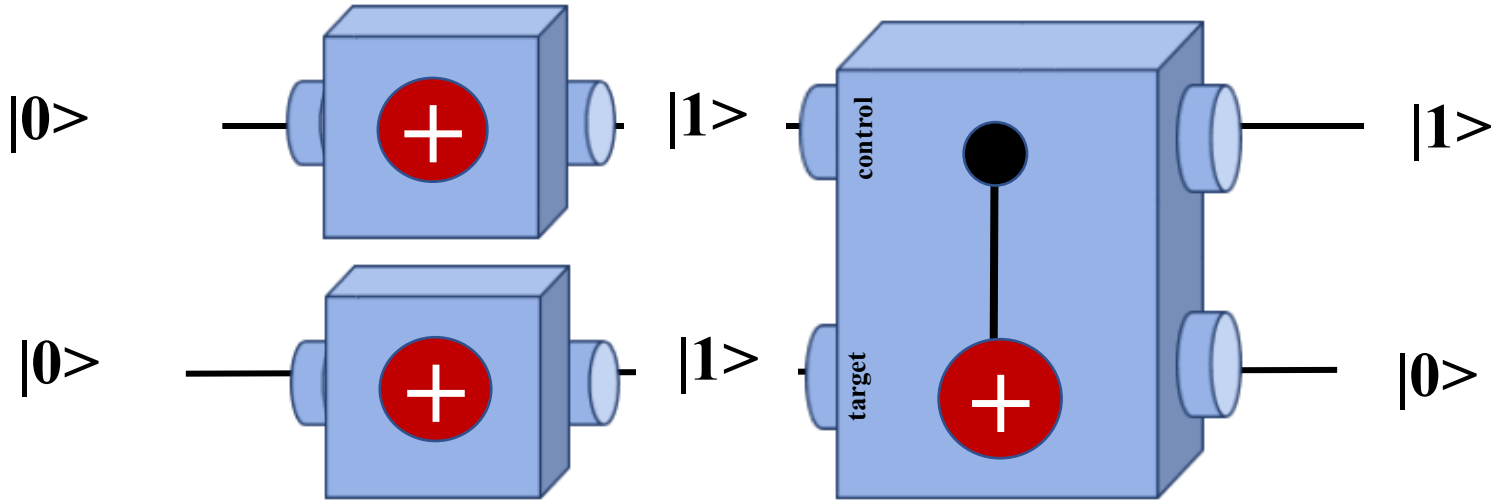
Have show numerical representation (on/off)



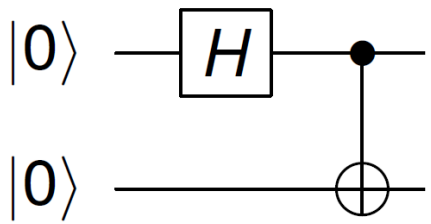
3)

Task: Find the output for the circuit

Have show numerical representation (on/off)



Entanglement: H+CNOT create



$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|00\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}\left[|0\rangle+|1\rangle\right]|0\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle+|10\rangle\right] \xrightarrow{CNOT} \frac{1}{\sqrt{2}}\left[|00\rangle+|11\rangle\right]$$

Quantum Computing: Elements of Quantum Circuit

- **Every computation has three elements: *data, operations, and results***



- **In *Quantum Circuit***

EDUCBA

Data ➔ Qubits

Operations ➔ Quantum Gates

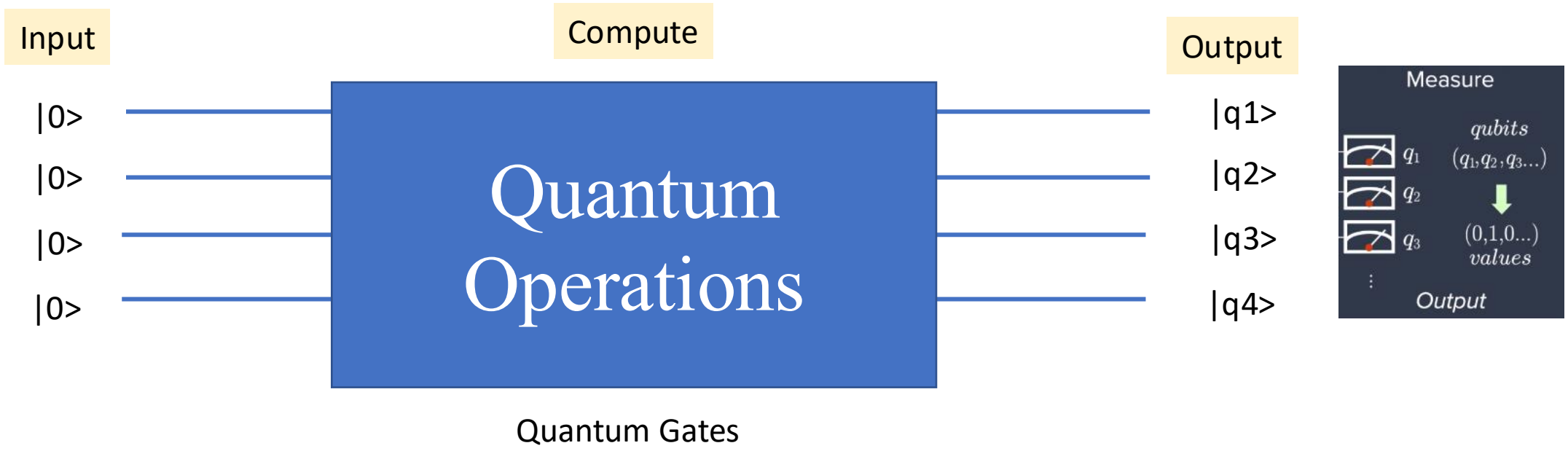
Results ➔ Measurements



Measuring the qubit state: collapsing the superposition state

Results = Measurements

Quantum Computation



Basics

• Measurement: consider it as a **readout operation** ➔ the system encodes state '0' and '1' ➔ we make a measurement to find out which one it is

• Quantum measurements are random ➔ probabilistic outcome: (a) with probability p_1, the output is '1' and (b) with probability $p_0 = 1 - p_1$, the output is '0'

• Before measurement: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ ➔ superposition between '0' and '1'

• After measurement: waveform collapse ➔ results is either $|0\rangle$ or $|1\rangle$

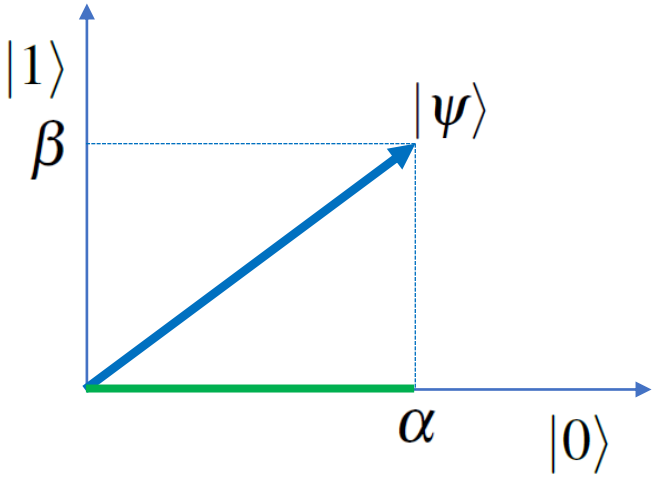
• Projective measurement: basis vecto $\mathscr{S} = \{|0\rangle, |1\rangle\}$

How much of '0'/'1' in my qubit?

Born Rule

• The probability that a state $|\psi\rangle$ collapses during a projective measurement onto the basis $\{|v\rangle, |v^{\perp}\rangle\}$ to the state $|v\rangle$ is given by

$$p(|v\rangle) = |\langle v|\psi\rangle|^2 \qquad \sum_j p(|v_j\rangle) = 1$$



Ex: $|\psi\rangle = \frac{1}{\sqrt{3}}\left(|0\rangle + \sqrt{2}|1\rangle\right)$ is measured in the basis $\{|0\rangle, |1\rangle\}$

bra ket

$$P(0) = |\langle 0|\psi\rangle|^2 = \left|\langle 0|\frac{1}{\sqrt{3}}\left(|0\rangle + \sqrt{2}|1\rangle\right)\right|^2 = \left|\frac{1}{\sqrt{3}}\underbrace{\langle 0|0\rangle}_{1} + \sqrt{\frac{2}{3}}\underbrace{\langle 0|1\rangle}_{0}\right|^2 = \frac{1}{3}$$

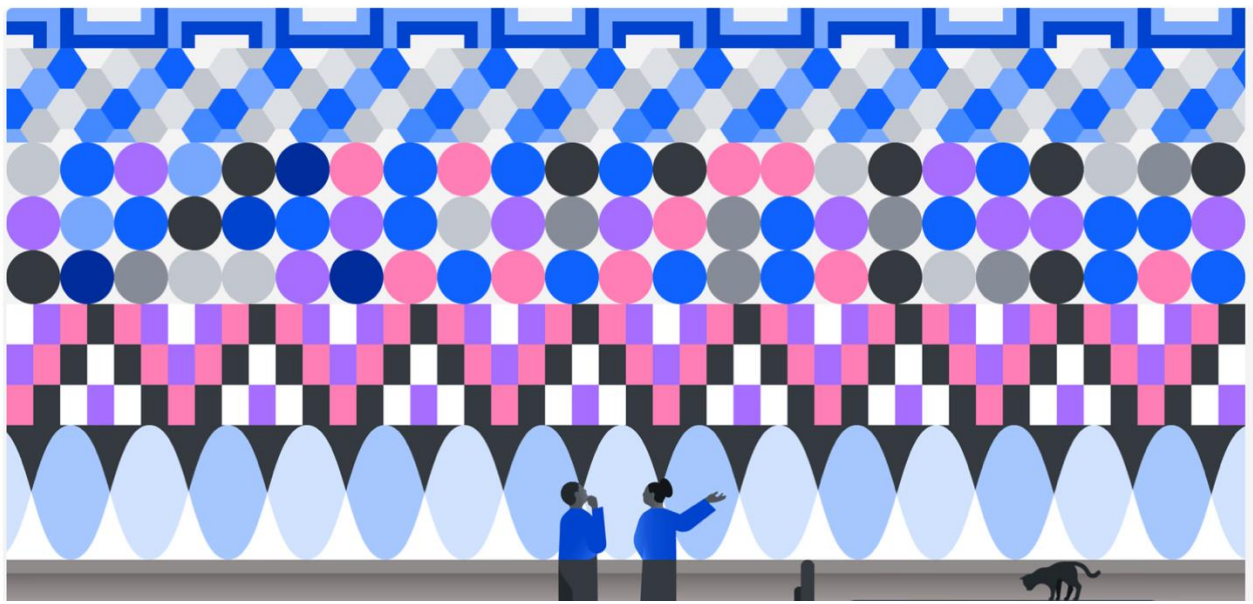
$$P(1) = 1 - P(0) = \frac{2}{3}$$

IBM Quantum Lab will be sunset on 15 May 2024. Learn more →

IBM Quantum Learning

Learn the basics of quantum computing, and how to use IBM Quantum services and systems to solve real-world problems.

Explore the latest course



Fundamentals of quantum algorithms

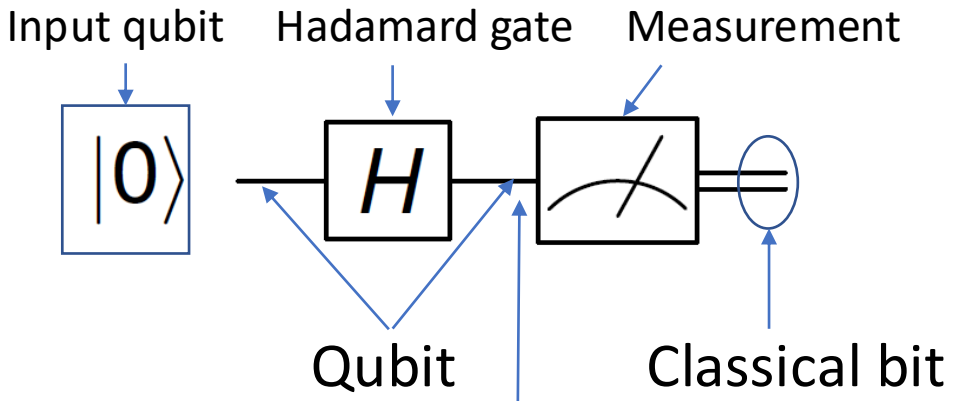
Video

Use quantum computers to solve problems more efficiently, including problems with real-world relevance such as searching and factoring.

Lessons Your progress

Hello, Quantum World!

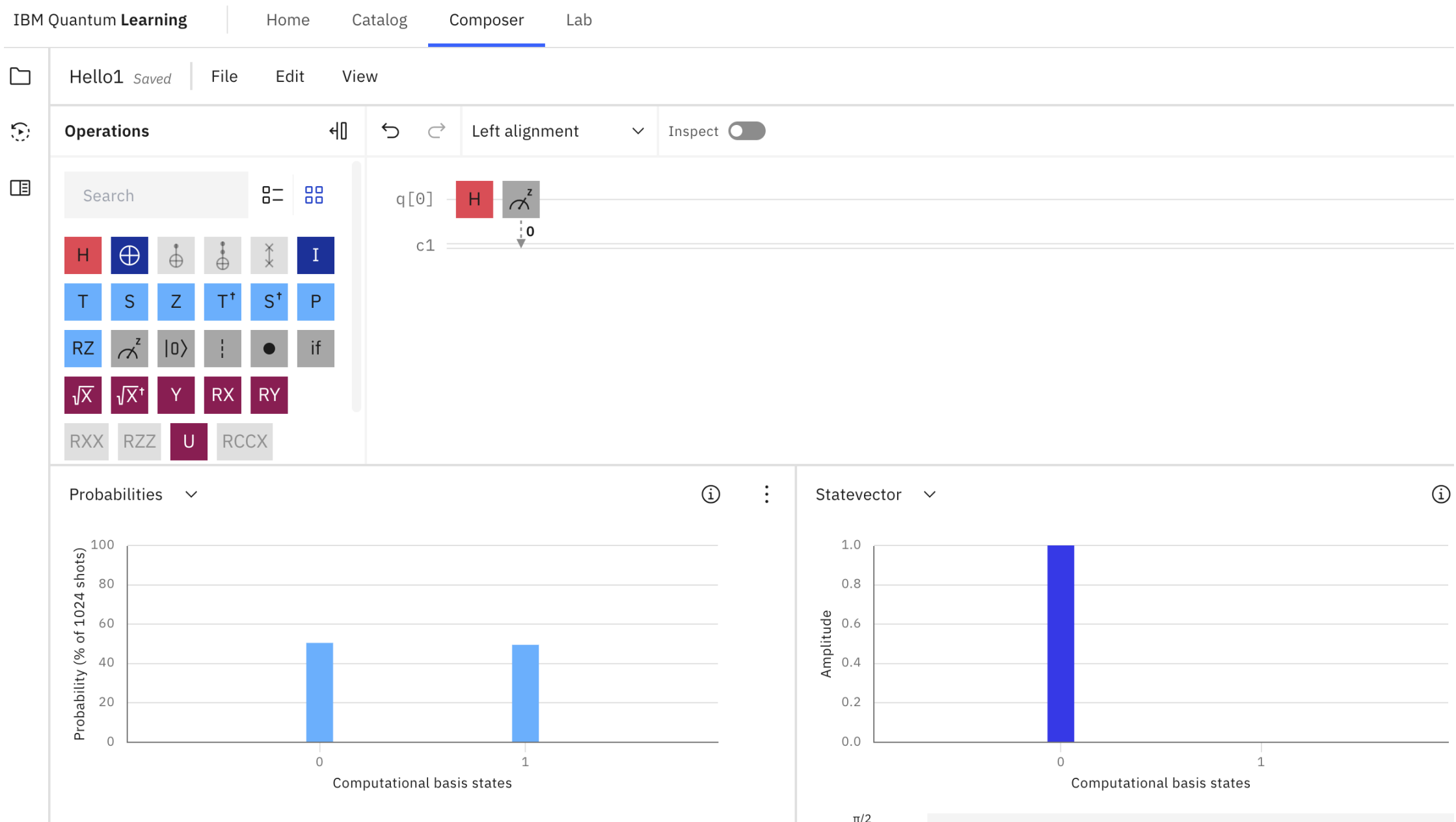
- Our first quantum circuit:



- After applying the H gate, we get: $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$
- After measurement, we get: 0 or 1 each with probability 50%

A circuit that generates perfectly random bits!

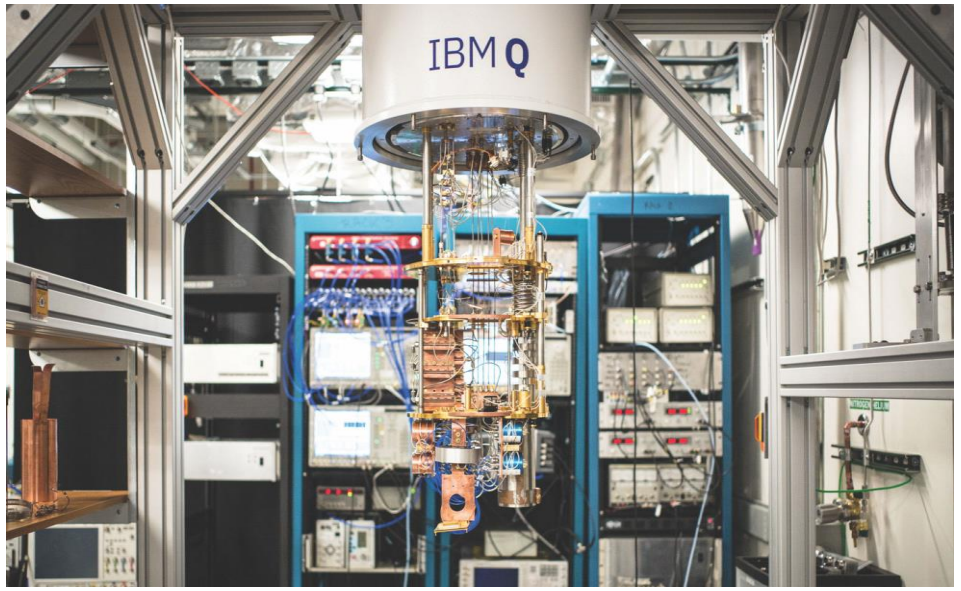
https://learning.quantum.ibm.com/



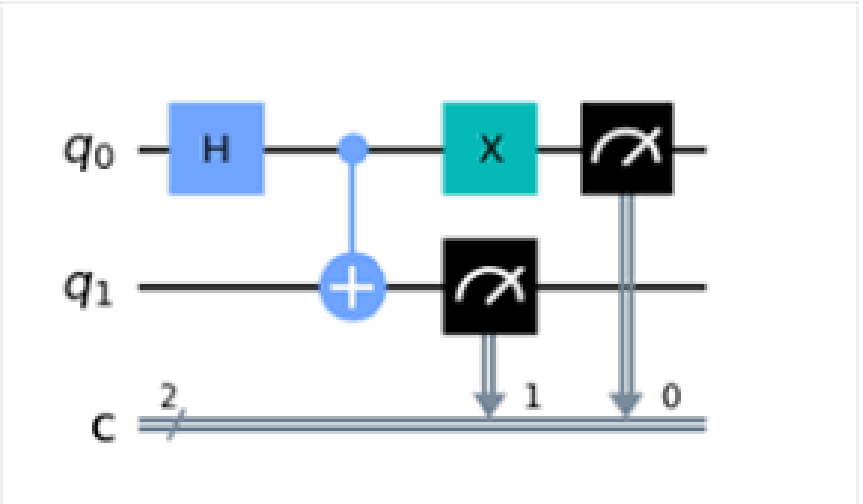
Coding on a Quantum Computer

https://quantum-computing.ibm.com

Qiskit







Using IBM Qiskit Lab a Python environment

QHack-intro-D1- Qiskit_Fa26

https://colab.research.google.com/drive/1Ex79sQHXKH525Stzs6eS1DqMsxKRsAID?usp=sharing

```
from qiskit import QuantumCircuit, transpile, assemble # To create quantum circuits
from qiskit_aer import Aer # Import Aer to simulate cicuit
from qiskit.visualization import plot_histogram  # To visualize the results


# Initialize the Quantum Circuit
qc = QuantumCircuit(1, 1)  # 3 qubits, 3 classical bits to store measurement results

# Apply gates
qc.h(0)
qc.x(0)

# Measurement operations to map the state of qubits to classical bits
qc.measure(0, 0)  # Measure qubit 0 and store the result in classical bit 0

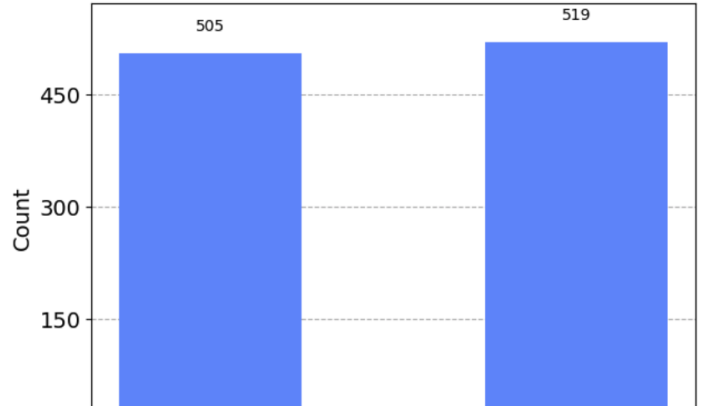

# Set up the backend and transpile the circuit for it
backend = Aer.get_backend('aer_simulator')
#tqc = transpile(qc, backend)

# Run the simulation
job = backend.run(qc,shots=1024)
result = job.result()

# Retrieve and print the counts
counts = result.get_counts()
print(counts)
plot_histogram(counts)

#qc.draw('mpl')
```

```
{'0': 505, '1': 519}
```



!pip install qiskit-aer

What is next

Take Home:
Lect2 and Notebook 2-more advanced, Important for the Hackathon challenge-**go through it on your own**

- https://colab.research.google.com/drive/10smA_4jod5bgF9tcoeMVcRjV1BlmpUQb?usp=sharing

- This notebook learning:

A quantum circuit prepares a state. We choose what to measure, run the circuit many times, and use the measurement outcomes to extract information about the state. This is also how quantum algorithms such as VQE estimate energy.

Next meiting Friday Oct 16, 2026

Next week: Training 2: From Foundations to Quantum Algorithms (VQE, QML, QAOA)

Presenters (Abhijit Chakraborty, David Liu, Pushpita Chatterjee, Dinh Nguyen, Lei Qian, Varun Puram)

Friday, October 16 | 12:00–1:30 PM CDT | Online (Zoom)

Training 2 will move toward Qiskit, VQE, and Quantum Machine Learning, providing more direct preparation for the hackathon challenges.